

Highlights

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

Emrehan Metin, Yunus Emre Cogurcu

- Scoring the same model at the fire, not region-wide, costs 0.143 AUC
- The cause is the negative pool, not prevalence: +0.155 against -0.000
- Under blocked CV, thermal predictors add +0.06 to +0.15 AUC in five regions
- Equalising the evaluation frame lifts mean transfer from 0.541 to 0.616
- None of 20 transferability diagnostics is shown to order transfer skill

Evaluation geometry and the limits of cross-region transfer in pre-fire thermal wildfire prediction

Emrehan Metin^a, Yunus Emre Cogurcu^{a,*}

^a*Çukurova University, Adana, Türkiye*

Abstract

Pre-fire thermal dryness separates a fire year from a normal one, but such models are rarely tested outside the region where they were fitted. Six thermal predictors were added to a terrain, fuel and greenness baseline in five Mediterranean regions, with MCD64A1 labels and spatially blocked validation. The first result is **evaluation geometry**: with model, predictors and fitting held fixed, scoring on the burn scar and its 2 km collar rather than region-wide costs **0.143 ROC-AUC** [**+0.077**, **+0.208**]. A control isolates the cause as the negative pool's composition, not class balance. Applied between regions the same effect withdrew five of our own claims: the study areas enclose very unequal far fields, and equalising them to a 10 km collar, which drops no burned cells, lifts mean transfer from 0.541 to 0.616 and leaves no between-region sign reversal supported under our own criterion, though the LST anomaly still differs on a difference interval. Local skill nonetheless does not travel. The thermal block adds +0.045 to +0.148 within regions at 5 km blocking, but +0.022 [−0.032, +0.077] when a whole burn scar is withheld and +0.004 [−0.028, +0.036] across twenty transfer directions, with a sign

*Corresponding author.

Email address: ycogurcu@cu.edu.tr (Yunus Emre Cogurcu)

that varies by pair. The static baseline transfers no better, at 0.537 against 0.541, and equalised transfer sits 0.155 below a matched within-region reference. Of twenty candidate diagnostics from five families, none was shown to order transfer, and the two that appeared to do so need target labels in both regions. Transfer skill has to be measured, not inferred from similarity.

Keywords: wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns (Pausas and Keeley, 2021). The dominant pattern in fire susceptibility mapping (Vibhandik et al., 2026; Jodhani et al., 2026) is settled: geospatial predictors are assembled over a study region, paired with a historical burned-area record and fitted with a supervised classifier, most often a random forest (Breiman, 2001), and the surface is published with a cross-validated AUC of 0.85 to 0.95. **That pattern does not report portability**, and this paper measures it.

1.1. *Portability goes unmeasured*

A susceptibility model’s reported skill is almost always a *within-region* estimate: held-out folds come from the same study area and season, often the same fire, and random folds over cells let spatial autocorrelation inflate it further. The problem is documented across ecological modelling (Roberts et al., 2017; Ploton et al., 2020) and addressed by spatially blocked cross-validation (Valavi et al., 2019; Meyer et al., 2018), but blocking does not make the estimate honest about a fire the model has not seen.

Because only that side of the ledger is reported, portability is never entered: a predictor block is adopted on the increment it delivers inside its training footprint, and whether that survives a change of region is not asked, even though any regional product built from locally trained models implicitly promises generalisation beyond it. Meteorological fire-danger indices are known not to port cleanly in a Peruvian case study (Podschwit et al., 2022), and the two studies that test transfer systematically (Dimarco et al., 2026; Liu and Mostafavi, 2025) both report that it largely succeeds between similar regions — and both transfer models whose dominant predictors are *spatially stationary*. Whether a dynamic, season-specific class behaves the same way is this paper’s question.

1.2. Pre-fire thermal dryness is the natural test case

Pre-fire thermal dryness is the dynamic class most plausibly *expected* to transfer. Standard predictors are static or near-static over the timescale at which fire danger varies, so they explain poorly why one summer burned and the preceding one did not; what changes is the state of the surface, to which satellite thermal observation gives partial access (Section 2.2). The physics linking moisture stress to combustion is universal, so portability should be most expected here, which makes the class diagnostic: a loss cannot be dismissed as a peculiarity of a locally defined covariate. The expectation is sharpest for the internally normalised channels, which should be least exposed to absolute-temperature offsets between regions; **they transfer no better than the absolute ones** (Appendix A(f)). The expectation motivates the design and is not a finding of it — Section 4.4 reports the associations running the other way.

1.3. *Why the loss is invisible to the diagnostics in use*

The field’s instrument for anticipating transfer failure is predictor-space
45 dissimilarity — the area of applicability and its relatives (Meyer and Pebesma,
2021, 2022; Ludwig et al., 2023) — which asks whether a target’s predictor
values fall inside the training envelope. That is a question about the marginal
distribution of the predictors, and it cannot see a change in the *relationship*
between predictors and response, because detecting one requires the labels
50 being withheld. If the failure is conditional rather than marginal, those di-
agnostics look in the wrong place; Section 4.6 tests whether they do.

1.4. *Contributions*

Three findings carry this paper, stated here as claims and established in
Section 4, which carries every interval.

55 **Contribution 1. Where a model is scored decides what it ap-
pears to know, and the effect is large enough to dissolve findings
of our own.** That evaluation extent inflates AUC is established in species
distribution modelling (Section 2.4); what is new is a magnitude under a
controlled design. Holding model, predictors and fitting fixed and changing
60 only which cells are scored costs **0.143 ROC-AUC**, the size of the incre-
ments this design measures for a predictor block, with the cause measured as
the negative pool’s composition rather than class balance (Section 4.3). Ap-
plied between regions it withdraws five claims of our own, including the sign
reversal we had offered as the mechanism and the one arm that held place
65 fixed (Section 4.4), and the practical consequence is a reporting standard
(Section 5.7).

Contribution 2. Local skill does not travel, and correcting the frame does not rescue it. The thermal block is worth a substantial within-region increment in all five regions, every bootstrap interval above zero, and stays positive when the frame is equalised — but much of it is a property of interleaved holdout, and across twenty transfer directions the paired contribution spans zero on both frames under the primary resampling unit, though much closer to the boundary once equalised and not under every admissible unit (Section 4.4), with a sign belonging to the pair rather than the block. Two controls bound this: the static baseline transfers no better on either frame, so the failure is not the thermal block’s peculiarity, and on matched frames and blocking equalised transfer still falls **0.155** short of the within-region reference. The within-region half is not novel (Alkan Akıncı and Akıncı, 2023; Iban and Sekertekin, 2022); the paired contrast against portability is.

Contribution 3. The shortfall cannot be anticipated by any diagnostic we could run. Of twenty candidates from five families, eighteen were not shown to order the matrix on the frames as drawn. Two conditional variants did, but both need burned labels on both sides, both rest on a data-selected feature subset whose all-nine counterparts span zero, and **Contribution 1 removes even those** (Section 4.4). The point survives without any ranking: at the point estimates the pair with the highest burned-niche overlap is among the weakest in the matrix while the lowest transfers above chance in both directions, contradicting an expectation the fire literature carries (Dimarco et al., 2026; Liu and Mostafavi, 2025) and agreeing with species distribution modelling (Vesk et al., 2021; Rousseau and Betts,

2022).

Two consequences follow, as supporting results: label-free alignment by standardisation and covariance alignment (Sun et al., 2016) does not repair
95 transfer, in what we believe is its first application to fire susceptibility; and because the residual is conditional, the resource that closes it is target labels, priced in Appendix A(u).

The leakage-audited, spatially blocked protocol is released with code, configuration and frozen outputs, so most of this can be re-run rather than
100 taken on trust, the three exceptions being named in the declarations — which matters given evidence that wildfire transfer conclusions are sensitive to evaluation design (Xu et al., 2026). A companion paper treats the observational layer beneath this one.

2. Related work

105 2.1. Fire susceptibility mapping, and what it reports

The dominant pattern is described in Section 1; the figure it publishes, a cross-validated AUC of 0.85 to 0.95, is a within-region estimate. Comparable within-region results already exist for the landscapes studied here (Alkan Akıncı and Akıncı, 2023; Iban and Sekertekin, 2022).

110 2.2. Pre-fire thermal dryness

Satellite thermal observation gives repeated access to surface state through fuel moisture content (Yebra et al., 2013). Chuvieco et al. (2004) established the pairing of land surface temperature with a vegetation index as a live-fuel-moisture estimator for fire-danger rating, and the Temperature-Vegetation

115 Dryness Index (Sandholt et al., 2002) formalises that feature space into an internally normalised measure, which is why it is often expected to travel better than raw temperature. That pre-fire thermal state carries information about subsequent fire is established (Maffei et al., 2018; Maffei and Menenti, 2019; Maffei et al., 2021), and Gelabert et al. (2025) found dead fine fuel moisture
120 and its anomalies the most influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been tested is whether such a classifier survives strict, label-free application to an unseen region.

2.3. Spatial validation and transferability

125 Random cross-validation over spatially autocorrelated cells inflates skill estimates (Roberts et al., 2017; Ploton et al., 2020), and spatially blocked designs are the standard remedy (Valavi et al., 2019; Meyer et al., 2018), though they have themselves been argued to introduce pessimistic bias (Wadoux et al., 2021; Milà et al., 2022; de Bruin et al., 2022). This paper blocks
130 throughout and reports every result at three block sizes, so a reader can see which verdicts depend on the choice.

Blocking corrects the inflation random folds produce within a region. It does not speak to performance on a fire the model has not seen, which Section 4.3 measures on identical cells. For that, the field’s instrument is
135 the area of applicability and related predictor-space dissimilarity measures (Meyer and Pebesma, 2021, 2022; Ludwig et al., 2023), which ask whether a target’s predictor values fall inside the training data’s envelope. Species distribution modelling has examined transferability far more systematically than fire has (Yates et al., 2018), and two studies there report that geographic

140 and environmental similarity do not reliably predict transfer success (Vesk
et al., 2021; Rousseau and Betts, 2022); transferability of satellite-derived
empirical models is an active question in fire remote sensing itself (Marino
et al., 2024).

The canonical taxonomy separates covariate shift, where the predictor
145 distribution moves but the predictor-response relationship holds, from con-
cept shift, where the relationship itself changes (Moreno-Torres et al., 2012).
Covariate shift is in principle correctable without target labels, by per-region
standardisation or covariance alignment such as CORAL (Sun et al., 2016),
and domain adaptation has an established remote-sensing literature (Tuia
150 et al., 2016; Persello and Bruzzone, 2012). Concept shift is not: no realign-
ment of inputs can repair a reversal in the sign of an association, because
detecting one requires the labels being withheld. Shift decomposition in ap-
plied remote sensing is not itself new (Huang et al., 2026), but we are aware
of no prior application of covariance alignment to fire susceptibility, fire oc-
155 currence or burned-area prediction.

2.4. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. (2022)
report, in a Peruvian case study, that meteorologically derived danger in-
dices do not port cleanly between fire environments. WildfireGenome (Liu
160 and Mostafavi, 2025) runs a leave-one-county-out matrix across seven US
counties and reports strong within-county performance with highly variable
off-diagonal transfer, similar pairs transferring well and dissimilar pairs col-
lapsing; its label is a principal-component composite of hazard *indicators*
rather than observed burned area, and it applies no adaptation. Xu et al.

165 (2026) argue that wildfire transfer conclusions depend strongly on evaluation design, a caution we address by fixing the protocol in a project log before the diagnostics were computed and by reporting every sensitivity axis; that log is not a formal pre-registration, and Section 3.11 states what was fixed and when. Kondylatos et al. (2023) provide Mesogeos, a 1 km Mediterranean
170 datacube.

Evaluation extent and AUC. Species distribution modelling settled long ago that the area a model is evaluated over is not a neutral choice. Lobo et al. (2008) make it the fifth and, by their own ranking, most important reason to distrust AUC comparatively: the extent of the modelled
175 area governs how many easy absences enter the calculation, and therefore the score. VanDerWal et al. (2009) show the same lever on the calibration side, and Barve et al. (2011) give the argument its general form as the accessible area. That literature is qualitative about magnitude, because magnitude is problem-specific, so our Contribution 1 is a measurement inside that frame-
180 work rather than a new phenomenon. The wildfire literature has largely not imported the lesson: region-wide figures are reported as though they described performance at the fire, and we know of no wildfire study that holds the model fixed and varies only the evaluation cells — which is why Section 5.7 treats the 0.143 as a reporting problem rather than a caveat.

185 **The nearest neighbour, and the contrast this paper draws.** Di-marco et al. (2026) is the closest Mediterranean analogue: 500 m predictors harmonised across four countries, tree ensembles under spatial cross-validation, transfer tested both leave-one-country-out and as a full 4×4 matrix, every transfer exceeding AUC 0.80 and bioclimatically similar coun-

190 tries scoring higher, with no domain adaptation.¹ Much of the design is shared — Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation, an explicit transfer matrix — and two things differ. **The predictor class:** their model rests on attributes of a place, all spatially stationary, ours on the state of a surface in one season.

195 **The response variable:** theirs is human-driven ignition, ours burned area, which have different dominant controls. Read together the two results suggest the relationship between domain similarity and transfer success may be predictor-class dependent, with Vesik et al. (2021) and Rousseau and Betts (2022) aligning from species distribution modelling while Dimarco et al. and

200 WildfireGenome run against it. We present that as a live disagreement; Section 5.6 states why our data do not settle it.

3. Data and methods

3.1. Study regions and temporal windows

Five Mediterranean wildfire regions are analysed (Fig. 1): Manavgat

205 2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangle in EPSG:4326, defined from place coverage rather than from a fire perimeter and deliberately not clipped to it, so that unburned cells around each fire form the negative class; none was tuned on burned prevalence, gate outcome

210 or any model metric. **One choice was label-informed and is stated as**

¹Their Results text refers to “LST anomalies”, inconsistent with their own Methods, where no land surface temperature or TVDI variable appears and the only temperature predictor is a static ERA5-Land seasonal climatology (Muñoz-Sabater et al., 2021). We follow their Methods and note the discrepancy so the comparison is transparent.

such: the North Evia box was extended after the legacy box proved atypically high in burned prevalence, the extended geometry then defined from place anchors and the legacy variant kept as a sensitivity arm. Section 4.4 shows this framing decision is consequential and Appendix C.5(x) treats it as the design lesson of the paper. A sixth region, Kozan 2023, is carried as a negative control and excluded by the gate of Section 3.3.

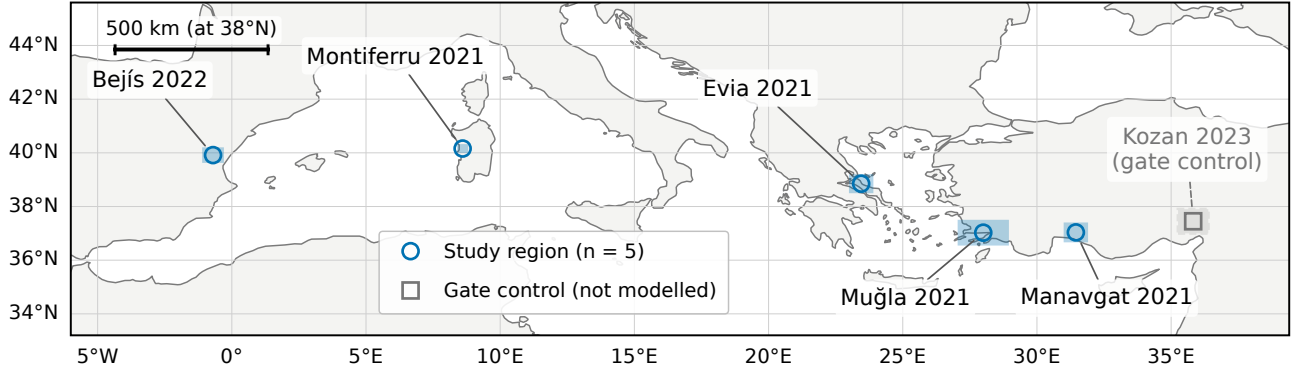


Figure 1: **Study regions and the gate negative control.** Locations of the five wildfire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table C1 to 5×10^{-3} degrees.

Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation can post-date the first labelled burning. Lengths vary with the event: 57 to

220 61 days for predictors, 35 to 59 for labels, counted inclusively. A four-year baseline of window-symmetric composites precedes each predictor window and supplies the climatological reference for the anomaly channels. Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and reproduced in Appendix C, Table C1.

225 3.2. Burned-area label and the ~ 500 m analysis grid

Labels come from MODIS MCD64A1 Collection 6.1 (Giglio et al., 2018) retrieved through Google Earth Engine (Gorelick et al., 2017), whose omission and commission characteristics (Boschetti et al., 2019) bound every model here. The analysis grid is **reconstructed rather than native**: the
230 pipeline’s 30 m EPSG:4326 reference grid is partitioned into 17 x 17 blocks, giving a nominal 510 m cell that approximates rather than reproduces the MODIS cell and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the label window; the label never affects eligibility for modelling. Two safe-
235 guards are recorded rather than assumed: cells burning inside a region’s own predictor window are removed, which ran for three of five regions, and burning in earlier years is screened for none. Appendix C.1 gives the full specification, including what follows from the grid’s shape.

3.3. Burned-landcover admissibility gate

240 Before any modelling each region passes a gate asking what fraction of its burned cells is dominated by natural vegetation, using ESA WorldCover classes on the same cells. A region is admitted at 0.50 with at least 30 burned

cells, and rejected as a cropland-dominated control at 0.50 cropland. Verdicts and the purpose of the gate are in Section 4.1, the full rule in Appendix C.1.

245 3.4. *Predictor variables*

Two feature sets are used. The **baseline** is terrain, fuel and green-
ness: elevation, slope, dominant land cover and a predictor-window median
vegetation-index composite, all static or near-static over the timescale at
which fire danger varies. The **thermal** set adds six pre-fire channels: current
250 land surface temperature, its anomaly against a four-year window-symmetric
baseline at the same cell, the Temperature-Vegetation Dryness Index and its
difference against that baseline, and two coordinate-informed products, a
downscaled and a fused surface temperature. The two differenced channels
are the ones constructed to isolate the dynamic anomaly. TVDI’s wet and
255 dry edges are percentiles of the values a given area and window happen to
contain, so **it is not portable as a physical quantity independently of
any concept shift** (Appendices C.4, C.5).

3.5. *Cell aggregation, validity and analysis populations*

Cell-level values are means over the ~ 510 m cell from valid 30 m pixels
260 only, with the valid fraction recorded; a cell is **valid_for_modeling** when
it meets a 30 % valid-pixel floor and no thermal channel is missing. The
primary population is natural vegetation, cells whose combined tree, shrub
and grass fraction reaches 0.50, which excludes cropland from every burnable
mask. The **secondary** population is all valid cells; the frozen export carries it
265 for the within-region arm in two regions, reported as a sensitivity in Appendix
A(v), and the transfer matrix is defined on the primary population only.

3.6. Classifier

The two feature sets of Section 3.4 are nested, the thermal set being the baseline plus the six thermal channels. Land cover is one-hot encoded and
270 numeric features are standardised inside the fitting pipeline. The classifier is a random forest (Breiman, 2001) with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced class weights and `random_state = 42`, identical for every region, population, feature set and transfer direction, so that no comparison here is confounded by a model choice.

275 3.7. Spatial-block cross-validation and bootstrap uncertainty

Cross-validation is spatially blocked. Blocks are squares of the analysis grid formed as `row_500m // B` by `col_500m // B`, and five folds are drawn over whole blocks, so no block is split between training and test. `B` is reported at 2, 10 and 20 cells, about 1, 5 and 10 km.

280 Uncertainty is a spatial-block bootstrap: blocks are resampled with replacement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the interval. Because it is blocks that are resampled, what bounds an interval's reliability is the number of blocks carrying at least one burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too few
285 such blocks it is stated as indicative rather than as an interval.

3.8. Cross-region transfer protocol

For each ordered pair the model is fitted on the source population and applied to the target with **no target label used at any point**: no refitting, no threshold selection, no calibration. Both feature sets are transferred, so
290 the thermal block's contribution is a paired difference within a direction

rather than a comparison across directions. All twenty ordered directions are computed.

3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets.

295 **Region-wise z-score.** Each region’s numeric features are standardised using its own statistics, source statistics from source data and target statistics from target data, never pooled. The classifier is refitted on the z-scored source and applied to the z-scored target. This removes first- and second-order marginal offsets.

300 **CORAL after region-wise z-score.** The source covariance is aligned to the target’s by the standard whitening-recolouring map (Sun et al., 2016), $\lambda = 10^{-5}$. Critically **the transform is applied to the source only**; the target is left as it is, and the classifier is refitted on the aligned source. Neither variant sees a target label, and both are verified label-blind at run time. λ
 305 sensitivity was assessed over nine values on four of the twenty directions, moving transfer AUC by at most 0.014; the canonical $\lambda = 1$ lies outside that sweep (Appendix A(b)).

3.10. Transfer-gap decomposition and the concept-shift criterion

The transfer gap is decomposed as (adapted – raw) / (within – raw),
 310 signed and unclipped, with its interval from the same paired bootstrap; Section 4.3 shows the remainder should not be read as a conditional residual, because much of it is incurred inside a single region (Appendices A(j), C.7).

The mechanism is diagnosed by **signed univariate association**: the raw ROC-AUC of each numeric predictor against **burned** is computed per

315 region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is a direction rather than weakness. A reversal is called bootstrap-supported only when the two regions' point estimates fall on opposite sides of 0.5 **and each region's own interval excludes 0.5**. That is stricter than requiring the intervals to be disjoint, and a feature failing the second condition is recorded
320 as a point reversal only.

3.11. *Transferability diagnostics versus transfer*

Twenty candidate diagnostics from five families are computed for every region pair and rank-correlated (Spearman) against observed raw transfer AUC, under one bootstrap that resamples unordered region pairs with both
325 of their ordered directions travelling together. The families are marginal predictor-space measures $P(x)$, including area-of-applicability dissimilarity, climatic and geographic distance and a learned domain classifier; burned-niche overlap $P(x|y=1)$; regime distance $P(y)$; and conditional direction agreement $P(y|x)$. **No family-wise error control is claimed**, and the
330 number of variants per family is reported with the result. Appendix C.2 gives the full specification.

3.12. *Interventions*

Pooled multi-region training. Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and
335 evaluated on the held-out region, with folds blocked as before. This asks whether pooling recovers what single-source transfer loses.

Removal of direction-reversing features. The two predictors reversing with bootstrap support **on the frames as drawn** are **elevation_**

mean and **lst_anomaly_mean**. Section 4.4 withdraws that support under
 340 an equalised frame, so the selection rule is frame-dependent and this arm is
 reported as a measurement under the original protocol, not as a consequence
 of an established reversal. Both are dropped and everything refitted, within-
 region and across every direction, and each is dropped singly so the cost can
 be attributed. The two features are selected by the same reversal analysis
 345 the result is then read against, so **both quantities are post-selection**
estimates with no correction applied.

3.13. Controls on the transfer path

Four evaluations establish what the transfer arms are measuring, all using
 the transfer protocol of Section 3.8 unchanged: a **within-region half-split**
 350 (modelled cells cut at the median of a grid axis, both axes and directions,
 a split discarded when either half is single-class); **leave-one-scar-out** (each
 burned component of at least 50 cells, dilated by a 2 km buffer, withheld
 from training and used as the target); a **foreign-region evaluation** of the
 same held-out scar areas; and the **same blocked model restricted** to those
 355 areas, which isolates the evaluation region from the training regime. Buffers
 of 2, 5 and 10 km were run, 2 km primary. The per-split and per-scar positive
 counts are unequal and bear on the interpretation (Appendix A(i)).

3.14. Leakage control and reproducibility

An explicit forbidden-column set is enforced at every model fit as an as-
 360 sertion rather than a convention: coordinates and their normalised forms,
 every burn-date and label-provenance column, and the agreement fraction
 are excluded from all feature sets. The natural-vegetation mask defines the

population and is never a predictor. All randomness uses seed 42 and the bootstrap 1000 replicates, with one qualification: the five diagnostic boot-
365 straps behind Table 5 and Table B1 use per-measure offsets from that seed rather than the seed itself, so that independent measures do not share a re-sampling draw. Two of the twenty transfer verdicts are not stable across seeds and are identified in Section 4.5. Because the transfer analysis runs in an environment separate from the upstream pipeline’s, every within-region model
370 was refitted against the frozen upstream output: the within-region comparisons agree exactly and the twenty transfer directions to within 1.6×10^{-7} , with the tolerance that applies if the library version is not pinned in Appendix C.5(vi). Headline results are repeated across two populations, three block sizes, the CORAL sweep, both feature sets and four classifier capacities,
375 and where a conclusion depends on one of those choices **the dependence is reported rather than resolved by choosing the favourable setting** (Appendices A, C.6).

3.15. Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla admits a comparison in which place is held fixed and the event
380 varies: a second fire burned inside the identical AOI, on the identical grid, eleven months after the first. Signed univariate AUCs are computed for both arms under the same 10-cell bootstrap used elsewhere. **Season, year and population all differ**, since the 2022 arm is defined by removing the 2021 scar (Appendices A(m), C.3).

385 4. Results

4.1. Study regions and the admissibility gate

All five candidate regions pass the burned-landcover gate and the negative control fails it as intended: Kozan 2023 returns a natural-vegetation fraction of 0.017 against the 0.50 threshold and is excluded. The separation is not marginal, the five admitted regions carrying 0.723 to 0.991, so
 390 the threshold falls in an empty interval rather than between neighbouring cases. That is consistent with the gate separating natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish, **though one negative control cannot establish it**. North Evia is analysed on an extended AOI, which leaves the burned scar essentially unchanged
 395 while cutting TSG prevalence from 0.676 to 0.287; the effect on transfer is in Appendix A(a).

4.2. Within-region: the thermal increment replicates in five regions

Adding the six thermal predictors to the baseline raises spatially blocked
 400 out-of-fold ROC-AUC in every region, the increment’s bootstrap interval excluding zero in all five at 1 km and at 5 km blocking (Fig. 2) — the two scales this design supports as intervals. At 10 km the point estimates hold, from +0.048 to +0.154, but rest on 6 to 33 positive-carrying blocks and are indicative.

405 *Table note (resampling units).* The bootstrap resamples spatial blocks, so an interval’s reliability is bounded by the number of blocks carrying at least one burned cell, and those counts fall from 192 to 843 at 2 cells to **6**

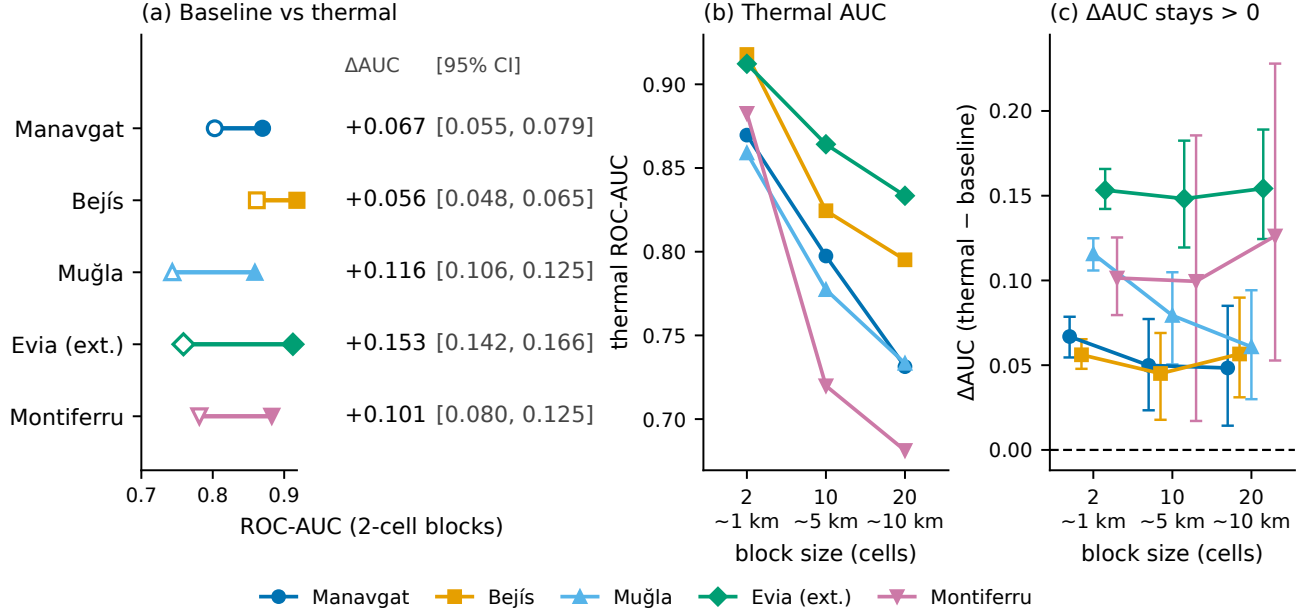


Figure 2: The thermal increment replicates in all five regions and survives coarser spatial blocking. Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired Δ AUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, $\approx$ 1, 5 and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment Δ AUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region-block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 1, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes (2, 10, 20 cells)**, and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

Table 1: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.7); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
2021	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
2022	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
2021	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
Evia					
2021					
(ext.)	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
2021	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

to **33** at 20 cells. An interval built on six such blocks has no meaningful coverage, so **the 20-cell row should be read as indicative rather than**
 410 **as an interval**. The 10-cell row is the coarsest blocking this design supports properly, every region there having 16 to 70 positive-carrying blocks, and the increment holds at that scale in all five regions (Appendix A(c)).

This within-region result is not itself novel; it is reported because the transfer arms below are measured against it.

415 4.3. *What the evaluation frame is worth, within one region*

This section isolates the effect Section 4.4 applies between regions. It comes first because it is measured on one model in one region with no transfer involved, so nothing the transfer arms do can explain it (Appendix A(i)).

Where the skill is lost, on a matched comparison. Four evaluations are reported; the last three are scored on **identical cells**, so they
 420 differ only in what the model was trained on, and the first shows why an unmatched comparison misleads. The held-out unit is a burned connected component of at least 50 cells with all cells within 2 km of it. What makes that harder than a whole region is the composition of its negatives, not its
 425 burned fraction, and that is measured rather than argued. Scoring the same predictions on a random sample of region cells drawn at the scar area's own burned fraction gives **0.782 against the region-wide 0.782**, so matching prevalence changes nothing, at $-0.000 [-0.003, +0.002]$; scoring them on the scar area gives 0.627, a fall of **+0.155 [+0.093, +0.217]**. That swaps both
 430 pools at once, so each was swapped singly: replacing only the negatives costs **0.147 [0.104, 0.191]**, the whole of it, and replacing only the positives costs **0.002 [-0.052, +0.048]**, null on average though the per-scar effect runs

from -0.12 to $+0.12$. **The effect is the negative pool** — every negative in a scar collar is fire-adjacent, sharing the terrain, land cover and synoptic conditions of the positives, whereas a region’s negatives include its easy far field — and ROC-AUC is in any case invariant to class balance at fixed class-conditional distributions (`pool_decomposition.json`).

Table 2: The four evaluations, scored on identical cells. Primary natural-vegetation population. Rows B, C and D are scored on the held-out scar area; row A is the whole region and is shown to make the mismatch visible. Means and Student t intervals are over the eight held-out scars.

Evaluation	Model trained on	Scored on	Mean AUC	95 % CI
A. Blocked cross-validation, 5 km	the region, scar included	the whole region	0.776	[0.738, 0.814]
B. Same blocked model, restricted	the region, scar included	the scar area	0.634	[0.552, 0.716]
C. Leave-one-scar-out	the region, scar withheld	the scar area	0.552	[0.501, 0.602]
D. Foreign region	another region, 306 to 2,802 km	the scar area	0.555	[0.495, 0.616]

All four rows are means over the **same eight scars**, and Bejís has no row C because it is its region’s only component of any size, which is why the controls above are computed on nine and report 0.782 and 0.627 against this table’s 0.776 and 0.634. The intervals are Student t over the scars, this arm’s resampling unit rather than the spatial-block bootstrap used elsewhere. Four scars are in Muğla and two in Montiferru, and **row A is a region-level quantity repeated identically across a region’s scars, so its interval is pseudo-replicated and should not be read as coverage**. That repetition propagates into $A - B$ and $A - C$; clustering by region gives $+0.155$ [$+0.082$, $+0.228$] and $+0.251$ [$+0.093$, $+0.409$], both still

excluding zero but the second 2.3 times wider, so **the scar-level intervals on these two differences overstate precision** (Appendix A(i)).

450 **The same model, scored two ways on the same region, differs by 0.143 AUC.** Rows A and B are one model and one set of out-of-fold predictions; the only change is which cells they are scored on. That change alone costs **0.143** [**+0.077, +0.208**] of the 0.225 [**+0.157, +0.293**] fall from A to C, about two thirds — a ratio of two estimates, compatible with anything
455 from roughly a third to nine tenths, so the point is the size of the numerator rather than the precision of the fraction. It is the size of the predictor-block increments this design itself measures, so a region-wide blocked figure is an upper bound on what the same model achieves where the fire actually is.

Neither withholding the fire nor moving 2,800 km has a measurable cost. B minus C, the fire-specific residual on identical cells, is **+0.082** [**−0.011, +0.175**]; C minus D, the effect of replacing a same-region model with one fitted 306 to 2,802 km away, is **−0.003** [**−0.075, +0.069**]. Both span zero and both arms sit close to chance, so **with eight scars, three of them starved of positives, this design cannot establish a fire-specific**
465 **residual, only bound it at about 0.18.** It is not shown to be zero on an unseen fire; what this design establishes is that it is not established there. Nor does it establish the fire *event* as the unit, since the held-out patch is defined by the labels and its identity cannot be separated from its location. Appendix A(z) reports the spread behind row D and the sweeps showing the
470 patch definition does not drive the result.

The increment declines with the holdout, and is not established once the fire is withheld. Contribution 1 is about the paired thermal-

minus-baseline difference, so the same evaluations were run on it: +0.056 to +0.153 under blocked cross-validation with every interval above zero, +0.027
 475 on a within-region half-split, **+0.022** [−0.032, **+0.077**] under leave-one-scar-out, and +0.004 [−0.028, +0.036] across regions. The point estimate falls monotonically as the holdout hardens and the last two intervals span zero, so **the within-region increment is substantially a property of interleaved holdout.**

480 *4.4. The same effect between regions, applied to our own matrix*

Section 4.3’s effect applies with equal force between regions. This test is reported before the transfer matrix because it changes what Section 4.5 and Appendix A(s) can claim; it withdraws nothing in Section 4.3, which is what it is built from. Sources are `aoi_frame_auc_frozen_mugla.csv` and `aoi_`
 485 `frame_transfer_frozen_mugla.csv`, which supersede the pre-correction files for the reason given below, with `collar_frame_bootstrap.csv` and `diagnostics_collar_frame.csv`; code is under `paper/code/` and the elaboration in Appendix A(w).

The five areas of interest are not comparable frames. Each is a
 490 rectangle drawn around a fire, and they differ by an order of magnitude in how much unburnt far field they enclose: the share of modelled cells beyond 10 km of any burned cell runs from **2.1** % in Montiferru to **63.1** % in Bejis (Appendix B, Table B6). That far field is not neutral — in Manavgat the median elevation of modelled cells rises from 472 m within 5 km of the fire
 495 to 1,273 m at 20 to 50 km, against 512 m for the burned cells themselves.

Under an equalised frame the sign reversals do not survive. Restricting every region to cells within 10 km of any burned cell removes only

far-field negatives — every burned cell is at distance zero and is retained at any radius — so the protection against a flattering radius is the sweep,
500 not the retention of positives. On that frame all five regions agree in sign on elevation, LST and TVDI, and both bootstrap-supported elevation reversals of Table B3 disappear. Under this paper’s own criterion from Section 3.10, **no between-region reversal remains bootstrap-supported:** the two features straddling 0.5 at the point estimate are the internally differ-
505 enced channels, and in each at least one region’s interval covers 0.5 — Evia’s by 0.003. The weaker difference instrument that Table B3’s note commits this paper to does support four opposite-sided pairs on `lst_anomaly_mean`, with three caveats that keep it weak. **The honest statement is that no between-region reversal meets the strict criterion under the collar,**
510 **and that the LST anomaly differs between regions on the weaker one.** Two scope limits belong with it, both in Appendix A(l) and A(w): the interval criterion was evaluated at the 10 km collar only, though at the point estimate the two radii agree exactly; and the collar reduces two regions’ cell counts substantially, so part of the loss of support is a loss of power, against
515 which features supported in both regions *rise* from 1.20 to 3.40 per direction.

The sign they now agree on is not the one the dryness framing predicts: at the point estimates a hotter pre-fire surface burned *less* in every region, and on mutual adjustment temperature survives where greenness does not, so the absolute channels behave here as static land-surface descriptors
520 rather than as dryness. Within distance to the nearest burned cell the sign crosses 0.5 in Manavgat and is attenuated to near-null in Montiferru, so in those two it is largely the residual spatial gradient the collar was introduced

to remove. The stratifications, the correlations and the reciprocal adjustment are in Appendix A(k).

525 **The same test determines what the diagnostics of Section 4.6**
can establish. Both candidates there with intervals excluding zero measure agreement in the sign of each predictor’s association between source and target — built from exactly the signed AUCs shown above to be frame artefacts, and correlated against transfer on the same unequal frames. Recomputing
530 both sides under the collar:

Table 3: The diagnostics that ordered transfer, recomputed on an equalised frame. Spearman ρ against target ROC-AUC over the ordered directions in which each is defined. The first two rows are the two variants that cleared zero in Table B1; the third is the all-feature cosine, which did not and is shown for contrast. Both sides are recomputed here under one bootstrap setting, 1000 replicates, seed 42. Source `diagnostics_collar_frame.csv` and `collar_increment_and_cosine.csv`.

Diagnostic	Full frame	10 km collar
Sign-agreement fraction, supported features	$\rho = +0.86$ ($p = 0.0001$, $n = 14$)	1.0 in all 18 directions, variance exactly 0 — degenerate
Cosine, supported features	$\rho = +0.81$ ($p = 0.0005$, $n = 14$)	$\rho = -0.06$ ($p = 0.82$, $n = 18$), variance 0.00014
Cosine, all nine features	$\rho = +0.50$ ($p = 0.023$, $n = 20$)	$\rho = +0.12$ ($p = 0.61$, $n = 20$)

The two fail differently and both fail: the agreement fraction has no variance left once every pair agrees, and the supported cosine keeps a trace of variance but stops tracking transfer. The disagreements they were reading were the far fields. **Only these two were recomputed, so the other**
535 **eighteen are unknown against the equalised transfer vector rather than shown to be null** (Appendix A(o)). This settles what Contribution

3 can claim: **no diagnostic tested here has been shown to order transfer**, and the two that appeared to were reading how the rectangles were drawn.

540 **Two further arms move with the frame.** The same-geography arm is the most extreme case in the cohort — a fixed study area is not a fixed evaluation frame, and its 2022 arm has 93.2 % of cells beyond 10 km of any burned cell against 55.3 % for 2021 — so under the collar its two arms fall on the same side of 0.5 on elevation, the feature whose reversal we had
545 reported. The claim is scoped: that arm is reconstructed from the 2021 predictors, valid for the year-invariant channels and **not** the seasonal ones, so **only elevation and slope carry a verdict, on both the reversal is removed, and on the thermal channels the arm is silent**; it rests on eleven positive-carrying blocks against a floor of sixteen (Appendices A(m),
550 A(o)). The within-region increment, by contrast, **survives**, positive in all five regions at a mean of +0.077 against +0.086 as drawn.

The transfer matrix moves as well. Restricting source and target to the same collar:

A data-provenance defect in this arm was found and corrected:
555 one region’s predictor file at the canonical path had come to differ from the one the frozen tables were computed on, so every arm here was re-run against the frozen export. It moves forty of a hundred per-direction values by up to 0.022 and leaves **every headline quantity above unchanged to within 0.0012**, with the signed AUCs of that region’s two channels moving by at
560 most 0.008 and no verdict changing (Appendix A(w)). The table reports the corrected values, and reproduces the frozen matrix at 0.541 against Table

Table 4: Cross-region transfer under equalised evaluation frames. Primary natural-vegetation population, thermal model, twenty ordered directions per row. Above/below chance are point counts; the supported counts use a 10-cell (≈ 5 km) spatial-block bootstrap on the target, 1000 replicates, seed 42; **Table B9 reports the same matrix under 2-cell (≈ 1 km) blocking**, which is why its supported counts are the larger 12 and 6 (Section 4.5). Per-direction bounds are in `aoi_frame_transfer_frozen_mugla.csv`.

Source frame	Target frame	Mean target AUC	Above chance	Below chance	Supported above / be- low	Paired thermal delta
full	full (Table B9)	0.541	14 of 20	6	9 / 4	+0.004
full	10 km	0.576	17 of 20	3	11 / 1	+0.003
10 km	full	0.570	17 of 20	3	11 / 3	+0.014
10 km	10 km	0.616	19 of 20	1	15 / 1	+0.023
5 km	5 km	0.608	19 of 20	1	12 / 0	+0.014

B9's 0.541 and 14 of 20 exactly. **The baseline control must be restated on this frame:** the static baseline transfers at 0.593 against 0.616, a paired difference of +0.023 rather than +0.004, so the control holds in kind but the gap is about six times larger once frames are comparable, and about four times at the 5 km collar.

What this settles, and what it leaves standing. Five quantities reported below are properties of the frames rather than of the predictor-burning relationship, and are identified as such where they appear: the count of six anti-predictive directions, which becomes one; the sign reversal of elevation, LST and TVDI as a mechanism; the sign-agreement diagnostic; the same-geography arm; and the paired thermal contribution, +0.004 as drawn against +0.023 equalised. That last quantity carries the portability null, so it is given an interval on the frame this section argues for: **+0.023 [−0.004, +0.048]** under the pair-cluster resampling of Appendix A(o). It still spans zero, so the null survives the correction — but only just, and under the alternative admissible unit, clustering by target region, it does not (Appendix A(w)). What survives is the central negative result, and its size must be stated on a matched comparison: setting 0.616 against a within-region reference of about 0.87 would compare a collar number with a full-rectangle one at 1 km blocking. Recomputed on the same frame at 5 km blocking that reference is 0.772, so the shortfall is **+0.155 [+0.094, +0.217]**, a Student t interval over the five target regions rather than the spatial-block bootstrap used elsewhere, and numerically almost identical to Section 4.3's negative-pool effect over nine scars, which is a different quantity on a different unit. It is real, and 0.155 rather than the 0.25 an unmatched comparison implies.

Appendix C.5(x) records the frame as a limitation of this cohort rather than of the method.

4.5. Cross-region transfer, and what label-free adaptation does to it

590 **Everything in this section is computed on the frames as drawn and should be read against Section 4.4,** which has shown the frames are not comparable and that the within-region reference used here is itself frame-dependent. The as-drawn matrix is reported because it is what the pre-registered protocol yields; per-direction values are in Appendix B, Table
595 B9, and the supporting arms in Appendix A(x).

Raw transfer is heterogeneous and includes anti-predictive directions. Target AUC spans 0.326 to 0.686. The support counts depend on the blocking scale, so both are reported: at 2-cell (≈ 1 km) blocking twelve of twenty directions are above chance with interval support and six below;
600 at the 10-cell (≈ 5 km) blocking this design defends, nine are above and four below with seven uncertain, the counts Table 4 uses. Of the below-chance directions only Manavgat to Bejís survives frame equalisation, and two verdicts are not stable across bootstrap seeds (`transfer_ci_blocksize.csv`). Even the best raw transfer sits far below that target’s own within-region
605 0.870, the raw deficit running 0.184 to 0.592 against a reference Section 4.3 shows is not matched to a transfer evaluation.

In precision terms it is worse than the ROC figures suggest. A susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity. PR-AUC averages **0.156 against a no-skill baseline**
610 **of 0.136,** and **six of twenty directions fall below their own baseline at the point estimate,** five of them with intervals entirely below it;

the exception is Bejis to Manavgat, whose interval covers its baseline. Only one direction exceeds twice its baseline. These are frame-as-drawn quantities and the PR arm was not recomputed on the collar (Appendix B, Table B4).

615 **The static baseline does not transfer either**, at a mean of **0.537** against **0.541** for the thermal model, so the failure is not specific to the dynamic block: a baseline that does not travel, plus pre-fire thermal state, gives a model that does not travel. Section 4.4 restates this control on the equalised frame, where it is 0.593 against 0.616.

620 **The thermal block’s paired contribution is +0.004, and the mean is not the informative statistic.** The individual contributions span **−0.148 to +0.133**, twelve positive and eight negative, so the spread is thirty times the mean and the sign belongs to the pair rather than to the block; a mean near zero records cancellation, not consistent absence of effect.
625 The directions are not independent, since each region appears in eight of the twenty, so the interval depends on the resampling unit — **all four units the design permits give the same answer**, from $[-0.027, +0.034]$ to $[-0.037, +0.046]$, and none propagates within-direction sampling variability. The leave-one-region-out jackknife shows how little the mean is anchored:
630 **dropping Evia alone reverses its sign.**

Label-blind adaptation compresses the matrix toward chance rather than repairing it. Under region-wise z-scoring the twenty directions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half the raw spread, and fourteen of twenty move closer to chance — which helps
635 the directions that failed and harms those that worked. The committed-in-advance CORAL arm averages **0.552**; taking whichever method scores

better per direction gives 0.556, but that selection uses the target labels the protocol forbids, so it is **an oracle upper bound rather than an achievable result**. Even the oracle only reaches the reference a model can reach on an unseen scar (Section 4.3), so alignment is regressing the matrix onto that reference rather than failing beneath it. What it cannot do is exceed the reference, and a sign reversal is not a distribution mismatch that realigning inputs would repair. Appendix A(x) reports the recovery fractions, including the seven directions with *negative* recovery.

645 4.6. *Transferability diagnostics: what appears to order transfer, and why it does not*

Twenty candidate diagnostics from five families were each rank-correlated with raw thermal transfer over the twenty ordered directions, under one common pair-based bootstrap.

650 **Only two diagnostics have intervals excluding zero, and both are conditional:** the sign-agreement fraction over interval-supported features at $\rho = +0.84$ $[+0.58, +0.88]$, and its cosine variant at $+0.81$. No marginal measure was shown to order the matrix, including area-of-applicability dissimilarity, climatic distance and geographic distance, and neither were the niche-overlap and regime families. The learned domain classifier is at ceiling, 655 separating source from target at $\text{AUC} \geq 0.96$ for every pair, so it always succeeds and carries no ordering information. Geographic separation is a single diagnostic here, centroid distance over the twelve directions for which the pipeline author’s export defines it, at $\rho = -0.24$ $[-0.84, +0.73]$: in the 660 expected direction and spanning zero.

Table 5: Transferability diagnostics versus raw thermal transfer, by family. Spearman ρ against raw transfer AUC with pair-based bootstrap 95 % CIs. Exp. is the sign expected if the diagnostic orders transfer. The member named is the one with the largest absolute correlation in its family, which is not always in the expected direction. All twenty individual diagnostics are in Appendix B, Table B1.

Family	Diagnostics	Exp.	Largest correlation in family	Sign as expected	Any CI excluding 0
P(y x) conditional	6	+	+0.84 [+0.58, +0.88] agreement fraction, supported features	yes	yes, 2 of 6
P(x y=1) niche overlap	5	+	+0.24 [−0.45, +0.74] Schoener’s D, 1-D mean	yes	no
P(x) marginal	6	−	−0.32 [−0.78, +0.33] domain-classifier AUC	yes	no
P(y) regime structure	2	−	+0.29 [−0.38, +0.74] log effective-N distance	no	no
geographic	1	−	−0.24 [−0.84, +0.73] centroid geodesic distance	yes	no

Four limits are stated with the result. **Size:** the index’s tie structure caps the achievable Spearman at +0.861, so the observed +0.840 sits on that ceiling, and its exact one-sided permutation p of 0.0060 — 240 of the $8! = 40,320$ relabellings of the **eight** pairs on which the index is defined — is the smallest that structure can produce, against a Bonferroni threshold of 0.05/19 = 0.0026. **No outcome of this diagnostic could have cleared family-wise correction.** **Labels:** signed associations need burned labels in both regions, so the family that appears to work is not available before deployment while the family that fails is. **Selection,** the sharpest: the two rows that clear zero are the *supported-feature* variants, whose subset is chosen by whether two regions’ bootstrap intervals happen to be disjoint — a data-dependent selection on the same data, with no correction; their unselected counterparts over all nine features are $\rho = +0.50$ $[-0.17, +0.83]$ and $+0.18$ $[-0.40, +0.72]$, both spanning zero, so **the result lives in the selection step.** **Frame,** established in Section 4.4, removes it entirely. Recomputing every row on the common twelve directions reproduces the published values to 4.8×10^{-5} and leaves the ordering unchanged, so it is not an artefact of unequal samples (Appendix B). Everything here is what the original protocol yields; the conclusion that survives is that **no diagnostic tested here was shown to order transfer once the frames are comparable.**

4.7. Further arms

Six further arms bear on the findings above without changing them: the contrast pair (A(s)), the two interventions (A(n)), the sensitivity summary (A(v)), the same-geography two-event arm (A(m)), the distance curve (A(t)) and the label-budget curve (A(u)). Each is stated there with its own limits.

The two bearing directly on Sections 5.4 and 5.7 are the interventions, where a local cost of -0.081 is measured against a transfer return of $+0.014$ $[-0.017, +0.045]$ whose interval spans zero, and the label budget, where thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s matched ceiling in
690 three of six directions and 30 to 57 % in the rest — 7 to 20 % of the target’s population, drawn from the event being predicted, which is not a resource available before that event burns.

5. Discussion

5.1. *Reading the three findings together*

695 Section 1.4 states the three findings and Section 4 establishes them; this section argues from them. One relation between them makes the paper cohere: the first is not a caveat attached to the other two, it is the instrument that sets their size. Applied to our own matrix it withdrew five claims, and what it left standing is a shortfall in transferred skill, not a reversed
700 relationship.

5.2. *Why the thermal increment is real but local*

The within-region increment and the transfer failure are measured at different separations, and Section 4.3 shows most of the difference is already present inside a single region. The increment is local in a specific sense: it
705 holds where held-out cells are interleaved with training cells and most of it is gone once they are not, before the fire or the region changes. It is not an artefact to be explained away, surviving every sensitivity arm of Appendix A(a)–(h), but it is established under interleaved validation and not beyond

it — and even there the window-closure arm **weakens monotonically in**
710 **Evia**, so what holds everywhere is survival, not improvement.

The natural objection is that Mediterranean regions are simply different systems, so a predictor meaning one thing in one place and another elsewhere is a comparison of two systems rather than instability. **We designed the two-Muğla-events arm to answer that objection and it does not**
715 **answer it**: Section 4.4 shows it is the most extreme frame artefact in the cohort. Three further confounds were never resolved in any case: season and year, which **cannot be resolved in this study area**, an unfixed population, and a positive-block count below this design’s own floor (Appendix C.5(ii)). With one fire per region everywhere else and that arm withdrawn,
720 **this cohort provides no evidence that the transfer shortfall is regional rather than event-specific**. Appendix A(t) adds a length scale for the within-region decay but cannot turn it into an attribution either.

One conclusion survives from the other direction: the static baseline transfers no better than the dynamic block, so whatever the shortfall is,
725 it is not the thermal block’s peculiarity.

5.3. Why the diagnostics fail, and what the conditional one is not

A diagnostic built on distance in predictor space asks whether the target’s predictor values look like the training data’s, which is not the question that matters when the failure is conditional: a target region can sit well inside
730 the training envelope while the relationship between those predictors and burning points the other way. Manavgat to Muğla is exactly that case, with 0.875 of target cells inside the full-frame area of applicability and transfer among the weakest in the matrix **on either frame**, while the least similar

pair transfers better (Appendix A(s)). High overlap does not buy transfer.

735 **The qualifier is not decorative:** at the 5 km blocking this paper otherwise defends neither contrasted direction carries a verdict, so this is a statement about point estimates. The domain classifier is at ceiling for every pair, so separability carries no ordering information either.

The two conditional variants that did clear zero are not a remedy. Section 4.6 states four limits with them: a tie-structure ceiling and a family-wise threshold **no outcome could have cleared**, a label requirement that makes it a mechanism diagnosis rather than a screen, a feature subset selected on the same data, and the frame, which removes the result altogether — the index is built from signed associations Section 4.4 shows to be frame artefacts. 740 So the practical conclusion is not “use this index instead”, nor even “it works but needs labels”: **none of the twenty candidates was shown to order transfer**, and the two that appeared to were reading how the rectangles were drawn — nulls on ten effective pairs, not shown to order transfer rather than shown incapable of it.

750 5.4. *What the two interventions do and do not show*

Both interventions show the same shape. Pooling four regions never beats the best single-source transfer for any target and stays well below the within-region ceiling (Appendix A(n)), so it does not manufacture the missing conditional information. Removing the reversing predictors costs -0.081 of within-region skill with interval support in every region, and returns $+0.014$ $[-0.017,$ 755 $+0.045]$ in mean transfer.

That pair of numbers is easy to read as an exchange, and it is not one. Both arms are null on the portability axis — the thermal block’s own con-

tribution is +0.004, removal returns +0.014, both intervals spanning zero —
760 so they measure a local cost and no compensating gain. That is not a con-
servation law and not a rate at which local skill can be sold for portability;
no such rate is estimated here.

5.5. *The regime hypothesis, reported as it happened*

A regime-structure explanation was stated in advance and the data con-
765 firmed the null: the regime-distance correlation has the wrong sign at the
point estimate, the most regime-similar pair fails in both directions, and the
most regime-different pair transfers above chance. One mundane explana-
tion can be set aside: subsampling Muğla, much the largest population, to
Manavgat’s cell count and separately to its positive count leaves its transfer
770 behaviour inside the subsampling range in both roles. The error was in the
hypothesised grouping, not in the data, and with ten pairs this cannot refute
regime typology (Archibald et al., 2013) in general.

5.6. *The empirical contrast with Dimarco et al.*

Dimarco et al. (2026) transfer successfully across a comparable Mediter-
775 ranean design and we do not, and the two results are not in conflict. Their
predictors are attributes of a place, ours the state of a surface in one season,
which suggests the relation between domain similarity and transfer success
is predictor-class dependent.

That reading is one of at least two, and we cannot separate them here:
780 their response variable is human-driven **ignition**, dominated by access and
activity, while ours is burned **area**, dominated by spread, so a predictor-
class explanation and a response-variable explanation are confounded in this

comparison. Our own data speak against a simple predictor-class reading in any case: the static baseline transfers at a mean of 0.537 here, so within this cohort the place-attribute class does not travel either.

5.7. Implications

In precision-recall terms, which is how a susceptibility surface is used, transferred models average a PR-AUC of 0.156 against a no-skill baseline of 0.136 **on the frames as drawn**, six of twenty falling below their own baseline (Section 4.5). Whatever the ROC figures suggest, a model moved to a region it was not fitted in does not usefully rank burned cells there.

The paper supports one concrete change in reporting: alongside a spatially blocked within-region figure, report skill on a held-out burn scar and its surroundings. On these five regions the two differ by about 0.14 ROC-AUC on the same model, the size of the effect such papers usually claim, so **a blocked figure alone should be read as an upper bound**. Transfer skill likewise has to be *measured* rather than inferred from similarity, since the diagnostics used for that inference did not order it here, and where a model must be moved the resource that closes the gap is target labels (Appendix A(u)): a real answer, but not a cheap one.

The results also bound what unsupervised alignment can be asked to do. Even the oracle selection, which uses the target labels the protocol forbids, only reaches the reference a model can reach on an unseen scar (Section 4.5). Alignment is not failing far below an achievable target; it is regressing the matrix onto it, at the cost of the directions that already worked. A sign reversal is not a distribution mismatch that realigning inputs repairs.

5.8. Limitations

Eleven limitations are stated in full in Appendix C.5; four bind the conclusions above. **The frames are not comparable and this cohort cannot**
810 **fully repair it** (Section 4.4): the collar radius is itself a choice — 5 km and 10 km give 0.608 against 0.616 — and the frames were fixed upstream, so we can restrict them but not extend them. Any future cohort should fix the frame by an explicit accessible-area rule (Barve et al., 2011) before any predictor is computed; that is the main design lesson of this paper. **Each region**
815 **contributes one fire season**, so regional concept shift is confounded with event meteorology and the shortfall cannot be attributed to region rather than event (Section 5.2). **The diagnostic correlations rest on an effective sample of ten region pairs**, so Section 4.6’s successes and failures both read at that power. And **the interval-support counts are less stable**
820 **than the point estimates behind them**: the published split turns on a lower bound of -0.00045 . The remaining seven, from the absent meteorological covariates to the single classifier family (Appendix A(h)), bear on scope rather than on the conclusions above.

6. Conclusions

825 Where a fire model is scored decides what it appears to know. Holding model, predictors and fitting fixed and changing only which cells are scored, moving from a whole study region to the burn scar and its 2 km collar costs **0.143 ROC-AUC**, with a control measuring the cause as the negative pool’s composition rather than class balance. That is the size of the predictor-
830 block increments this design itself measures, so a region-wide validation figure

should be read as an upper bound on what a model achieves where fire actually occurs, and reported as one.

That correction is not only other people’s problem. Applied to our own five-region matrix it withdraws five of our claims. The study areas enclose
835 very unequal far fields, and equalising them to a 10 km collar lifts mean transfer from 0.541 to 0.616, reduces the directions below chance, and leaves no sign reversal supported between regions under our own criterion. The same correction applies to the one arm that held place fixed, two fires in one study area eleven months apart, which we had wrongly exempted because
840 its geography was constant while its evaluation frame was not: its elevation reversal does not survive equalisation either, and that arm can speak only for the year-invariant channels. What had looked like a reversed predictor-burning relationship was mostly a statement about how five rectangles were drawn.

845 What survives is the central result. Pre-fire thermal dryness adds a real and repeatable within-region increment in all five regions, surviving spatial blocking at about 5 km and a predictor window closed up to two weeks before the first labelled burning. It does not travel: on matched frames and blocking, equalised transfer of 0.616 falls 0.155 short of the within-region reference;
850 the paired cross-region contribution spans zero on both frames under the resampling unit we treat as primary, though not under every admissible one (Section 4.4), with a sign that varies by pair; and the static baseline transfers no better on either frame, so this is not a peculiarity of thermal predictors. The shortfall also cannot be anticipated: none of twenty diagnostics from
855 five families was shown to order the matrix, and the two that appeared to

require target labels on both sides and stop ordering transfer once the frames are comparable. Label-free alignment compresses most directions towards chance rather than repairing them, and supervised recalibration works but is not cheap.

860 Two limits belong with all of this. The agreed thermal direction is the opposite of the one the dryness framing predicts — hotter pre-fire surfaces burned less in every region, and holding greenness does not remove it while holding temperature reverses greenness in two regions — so on this cohort the absolute channels behave as static land-surface descriptors rather than
865 as a dryness index. And with one fire per region, and the single arm that held place fixed now withdrawn, this design measures a transfer shortfall but cannot attribute it to region rather than to event.

Two practical consequences follow. Report local skill and portability separately, since a predictor block worth a large within-region increment may
870 contribute nothing distinguishable from zero across regions. And fix the evaluation frame by an explicit accessible-area rule before any predictor is computed, because a frame chosen for convenience can manufacture both a transfer failure and a mechanism to explain it. Three extensions are left for future work: temporal transfer, meteorological covariates, and physically
875 normalised dryness variables.

Appendix A. Appendix A. Sensitivity analyses

Each arm below varies one design choice and leaves everything else fixed. None changes a conclusion in the main text. They are reported so that a reader can see which choices were tested, and what each was worth. Arm (f)

880 is the exception in one respect: it is not a robustness check but a test of a claim the introduction makes, and the claim is not upheld.

Appendix A.1. $A(a)$ – $A(h)$. The eight sensitivity arms

Each arm is labelled by its letter below, and is cited elsewhere in the paper as Appendix A(a) through A(h).

885 **(a) Evia AOI and prevalence.** The raw transfer arms were repeated with the legacy, high-prevalence Evia box. Every qualitative conclusion is unchanged. Thermal raw transfer AUCs move by up to 0.07, the largest being Evia to Bejís at 0.378 to 0.448, and no direction changes side of the chance line.

890 **(b) CORAL regularisation, including the canonical $\lambda = 1$.** Over nine λ values from 0 to 10^{-1} on four directions, CORAL transfer AUC moves by at most 0.014 within any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ of the cited method lies outside that sweep and was computed separately on the same four directions. It changes the thermal
895 mean from 0.519 to 0.522, a shift of +0.003 with a per-direction range of -0.005 to $+0.016$, and none of those four directions crosses the chance line. The $\lambda = 1$ values are 0.510, 0.444, 0.559 and 0.575 against 0.508, 0.445, 0.564 and 0.559 at $\lambda = 0.1$.

Those four directions are Bejís–Muğla and Manavgat–Muğla,
900 **and they exclude the pair that moves.** An earlier version of this appendix generalised from them to the whole matrix, which was an error: the Manavgat–Bejís pair is not in the sweep and it is the pair whose CORAL result is λ -dependent. A separate frozen artefact, `experiments/cross_region/step10/coral_lambda_sensitivity.csv`, carries it. Bejís to Man-

905 avgat gives 0.5571 [0.5292, 0.5857] at $\lambda = 10^{-5}$, 0.5644 at $\lambda = 10^{-3}$ and 0.5473 at $\lambda = 0.1$, all with intervals entirely above 0.5, but **0.4850 [0.4496, 0.5197] at $\lambda = 1$** , which crosses the chance line and loses its interval support. Manavgat to Bejís falls likewise, from 0.5108 to 0.4776 [0.4501, 0.5013].

The correct statement is therefore narrower. CORAL-dependent conclusions are insensitive to λ over three orders of magnitude below 0.1; at the
910 cited method’s canonical $\lambda = 1$ the one direction whose adapted interval sits entirely above chance loses that status. $\lambda = 10^{-5}$ is the minimally regularised choice and matches the reference implementation, but the dependence is real and is reported rather than absorbed.

915 **(c) Blocking scale.** Recomputing the transfer quantities at 10-cell (≈ 5 km) blocking from the frozen per-cell predictions widens the intervals and moves the verdict counts, from ten positive, seven negative and three uncertain at 1 km to six, four and ten at 5 km. Coarser blocking therefore removes support from seven verdicts and adds none. Across five seeds the counts run
920 5 to 6 positive, 3 to 4 negative and 10 to 11 uncertain, so they are not exact.

The point estimates are unchanged. That is an identity rather than a result, and it should not be offered as robustness. The blocking scale is the bootstrap *resampling unit*, and each point estimate is computed once over all target cells, so no choice of block size could have moved one. The
925 comparison does establish two things. Every verdict that changes, changes towards “no verdict”. And no direction crosses the chance line under the widened intervals. The counts are the fragile part of this paper. The sign pattern is not thereby shown to be robust. It is simply not tested by this variation.

930 **(d) Predictor-window closure.** The predictor window was closed 7
and 14 days earlier in all five regions. The thermal contribution stays positive
and bootstrap-supported everywhere. The direction of the change is region-
specific. It strengthens in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla,
from 0.115 to 0.128. It is flat in Montiferru. It weakens monotonically in
935 Evia, from 0.156 to 0.149 to 0.135. What holds in every region is survival,
not improvement. That is the claim carried in Section 5.2.

(e) Quality screening of the coarse thermal input. Two of the
five regions' MODIS inputs are quality-screened and three are not. The
split follows export date rather than design, and it induces an elevation-
940 correlated change at the input, at $r = +0.615$ in Manavgat. Manavgat's
entire downstream chain was rebuilt from a quality-screened input. That
changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C.
No signed univariate association moves by more than +0.0003. Elevation
is identical in both arms, at 0.374, the population is unchanged, and the
945 within-region increment moves from $[+0.055, +0.079]$ to $[+0.054, +0.077]$.
The reason is structural. Elevation is a DEM variable the screening cannot
touch, and fusion falls back on the MODIS-derived surface across only 2.14
percentage points of coverage. Details are in `paper/modis_qc_downstream_
propagation.md`.

950 **(f) Normalised against absolute dryness channels.** Section 1.2
argues that an internally normalised index should be less exposed to absolute-
temperature offsets between regions than raw land surface temperature. The
thermal block contains both kinds, so the argument can be tested directly as
a feature-set contrast.

955 One thing should be said before the result, because it makes the outcome less surprising than it might otherwise appear. The normalised channels are the two that hold their direction in the same-geography two-event comparison of Appendix A(m), where the absolute channels move. That is a statement about one region across two fires. **Across regions it does not**
960 **hold: `lst_anomaly_mean`** is itself one of the two predictors whose reversal is bootstrap-supported on the frames as drawn, reversing between Bejís and Evia (Section 4.4 withdraws that support under an equalised frame), which is why Appendix A(n) drops it alongside elevation. Being internally normalised protects a channel against the offset between two seasons in one place. It
965 does not, on this evidence, protect it against a change of place.

Three feature sets were run over all twenty directions and all five within-region folds, with the classifier, population, folds and bootstrap held fixed. The harness aborts unless its reference configuration lands on the frozen exports. It did exactly: the maximum absolute difference between the reference
970 arm and the frozen transfer AUCs is 0.000000 across all twenty directions.

Feature set	Mean transfer AUC	Directions > 0.5	Mean within-region AUC
baseline only	0.5371	16	0.7896
baseline + normalised anomalies	0.5442	15	0.8551
baseline + absolute surface state	0.5479	13	0.8564
all ten features (reference)	0.5414	14	0.8883

The advantage is not found. Per direction, the normalised set minus the absolute set has a mean of -0.0037 . It is positive in 12 of 20 directions and runs from -0.099 to $+0.094$, so the spread is an order of magnitude larger than the difference. The normalised channels transfer very slightly worse at the mean, and nothing here supports a design rule favouring anomaly-referenced dryness for portability.

Within region the two sub-blocks are indistinguishable, at 0.8551 and 0.8564, and each recovers most of the gap between the baseline’s 0.7896 and the full model’s 0.8883. What separates them in sign stability does not become transferable skill.

This arm strengthens the negative finding rather than softening it. Four differently constituted feature sets were tried. None clears a mean of 0.548 across regions, and all differ sharply within them. The baseline arm also gives an independent confirmation of the control reported in Section 4.5: its

mean transfer of 0.5371 was recomputed here from the modelling datasets and matches the 0.537 read from the frozen per-direction export. Details are in `paper/anomaly_only_transfer.md`.

The comparison is between two sub-blocks of one thermal set on one cohort. It does not test normalised dryness indices in general, and it does not test a normalisation fitted against a pooled multi-region reference rather than each region's own baseline years.

(g) **The coordinate-informed channels.** `downscaled_1st_mean` and `fused_1st_mean` come from a per-region downscaling model whose own inputs include coordinates, which Section 3.14 names as the one route by which a coordinate-derived surface re-enters a feature set that excludes coordinates. A coordinate-smoothed surface is by construction locally informative and non-portable, so if the within-region increment depended on it, that increment would be this paper's own result in miniature rather than a finding about thermal dryness. The arm was therefore run here rather than deferred.

Both channels were dropped and everything refitted, over all twenty directions and all five within-region folds. The reference configuration reproduces the frozen exports exactly, at 0.000000 against both the transfer and the within-region references.

Region	Increment, full set	Increment without the two channels	Retained
Manavgat 2021	+0.067	+0.063	94 %
Bejís 2022	+0.056	+0.046	82 %
Muğla 2021	+0.116	+0.097	84 %
North Evia 2021	+0.153	+0.145	94 %
Montiferru 2021	+0.101	+0.105	103 %
Mean	+0.099	+0.091	92 %

The increment does not depend on them. It stays positive in every region and retains 82 % to 103 % of its size, rising slightly in one region. Mean transfer is likewise unchanged, at 0.545 without the two channels against 0.541 with them and 0.537 for the baseline alone. The local skill this paper reports is therefore not an artefact of a coordinate-smoothed surface.

This arm was computed independently here and reproduces the companion paper’s figure for the same quantity, 82 % to 103 %, from a separately written harness.

(h) Model capacity. Every number in this paper comes from one random forest with unlimited depth and `min_samples_leaf = 3`. That is the configuration most able to encode local structure and least able to extrapolate, so a reader may reasonably ask whether the transfer failure is a property of the predictors or of the estimator. Holding the model fixed is right for the internal comparisons and does not license the external claim, so three further estimators were run over the same twenty directions and the same five within-region folds.

Estimator	Within-region AUC	Within increment	Transfer AUC	Transfer increment	Above chance
Random for- est, depth un- lim- ited, leaf 3	0.888	+0.099	0.541	+0.004	14 of 20
Random for- est, depth 6, leaf 50	0.829	+0.055	0.556	−0.007	14 of 20
Random for- est, leaf 200	0.805	+0.047	0.550	−0.021	14 of 20
Penalised lo- gis- tic re- gres- sion	0.741	+0.045	0.510	−0.024	14 of 20

Regularisation does not rescue transfer. All four estimators land

1025 between 0.510 and 0.556, all four put exactly fourteen of twenty directions
above chance, and the linear model, the one built to extrapolate, transfers
worst. Within-region skill falls as capacity is reduced, from 0.888 to 0.741,
which is what regularisation is expected to cost; transfer does not rise to
meet it. The thermal block’s contribution to transfer is +0.004 under the
1030 canonical forest and **negative** under all three regularised alternatives, so the
block does not become portable when the model is made simpler.

The transfer failure is therefore a property of the predictors on this cohort,
not of an unregularised forest. Detail in `paper/model_capacity.json`.

Appendix A.2. A(i). The four evaluations of Section 4.3, in full

1035 Section 4.3 reports four evaluations of the same models as a ladder. The
per-split and per-scar detail is here, so that the ladder can be checked without
leaving the manuscript.

Table A1. Within-region half-split, every split. Source and target
positive counts are given because they are unequal, which is the principal
1040 limit on this arm: a straight cut does not produce two exchangeable halves.
Two splits are unusable because one half of Manavgat contains no burned
cells.

Region	Axis	Direction	Source positives	Target positives	Thermal AUC	Baseline AUC
manavgat 2021	east-west	low to high	700	84	0.695	0.661
manavgat 2021	east-west	high to low	84	700	0.557	0.580
manavgat 2021	north-south	low to high	0	784	—	—
manavgat 2021	north-south	high to low	784	0	—	—
bejis 2022	east-west	low to high	640	460	0.750	0.784
bejis 2022	east-west	high to low	460	640	0.578	0.560
bejis 2022	north-south	low to high	357	743	0.713	0.662
bejis 2022	north-south	high to low	743	357	0.603	0.613
mugla 2021	east-west	low to high	1,449	1,462	0.448	0.378
mugla 2021	east-west	high to low	1,462	1,449	0.636	0.628
mugla 2021	north-south	low to high	774	2,137	0.533	0.503
mugla 2021	north-south	high to low	2,137	774	0.294	0.549
evia 2021 extended	east-west	low to high	2,052	612	0.589	0.506
evia 2021 extended	east-west	high to low	612	2,052	0.645	0.482
evia 2021 extended	north-south	low to high	2,564	100	0.649	0.513
evia 2021 extended	north-south	high to low	100	2,564	0.533	0.495
montiferru 2021	east-west	low to high	410	129	0.475	0.461
montiferru 2021	east-west	high to low	129	410	0.507	0.513
montiferru 2021	north-south	low to high	438	101	0.583	0.511
montiferru 2021	north-south	high to low	101	438	0.536	0.447

1045 **Table A2. Leave-one-scar-out at a 2 km buffer, every scar.** Muğla
and Montiferru are the only regions containing more than one burned compo-
nent of at least 50 cells. Muğla is the only region where holding one out still
leaves the source model properly trained across every arm; Montiferru’s two
components are very unequal, so one of its arms retains 472 source positives
1050 and the other 97. Muğla’s four arms mean 0.579; the four arms in the other
regions mean 0.525, and the pooled figure of 0.552 averages the two.

Region	Component	Source	Target positives	Target cells	AUC
		positives left			
evia 2021 extended	1	11	2,653	3,059	0.465
manavgat 2021	1	88	696	1,135	0.592
montiferru 2021	1	97	442	758	0.584
montiferru 2021	5	472	67	195	0.458
mugla 2021	6	1,997	914	1,266	0.595
mugla 2021	1	2,173	738	1,244	0.561
mugla 2021	10	2,272	639	940	0.616
mugla 2021	8	2,363	548	954	0.542

The evaluation populations of the two arms are not comparable with each other or with the transfer targets. A held-out scar with its 2 km collar contains only fire-adjacent negatives, against a whole target region’s inclusion of its easy far field; the burned fractions, 34 to 87 % against 3.8 to 28.7 %, are a symptom of that rather than the cause, since ROC-AUC is invariant to class balance at fixed class-conditional distributions. The scar arm therefore asks for discrimination against the nearest and most similar negatives only, while a transfer arm includes the whole easy far field. Section 4.3 states the consequence: the last three rows of the ladder are not distinguishable by this design.

Table A3. The foreign-region arm, decomposed by source. Each held-out scar area is scored with a model fitted on each of the other four regions. Table 2’s row D is the mean over the **eight** scars that carry a row C, that is 32 of the 36 combinations below; the nine-scar mean quoted in this appendix is 0.559 against row D’s 0.555.

Target region	Scar	Mean over sources	Min	Max	Spread
Bejís 2022	1	0.589	0.521	0.631	0.110
North Evia 2021	1	0.590	0.374	0.704	0.329
Manavgat 2021	1	0.468	0.407	0.563	0.156
Montiferru 2021	1	0.504	0.457	0.546	0.089
Montiferru 2021	5	0.490	0.399	0.656	0.257
Muğla 2021	1	0.518	0.425	0.608	0.183
Muğla 2021	6	0.568	0.533	0.597	0.064
Muğla 2021	8	0.632	0.570	0.724	0.155
Muğla 2021	10	0.671	0.579	0.716	0.137

Over all 36 combinations here the mean is 0.559, the range 0.374 to 0.724,
1070 and ten fall below chance. Restricted to the 32 combinations behind row D
the mean is 0.555 and the same ten fall below chance. The mean spread
across the four sources for a single scar is 0.164 over the nine scars and 0.171
over the eight. Averaging over sources is what makes row D comparable with
row C, which is fitted on one region; it is not a claim that the choice of
1075 foreign source is immaterial, and Section 4.3 states the distinction.

**Table A4. Prevalence is not the cause of the evaluation-area
effect.** The same fitted model and the same out-of-fold predictions are scored
three ways: on the whole region, on a random sample of region cells drawn
at the scar area’s own burned fraction, and on the scar area. Twenty draws
1080 per scar, seed 42. Nine scars, since this control needs no leave-one-scar-out
arm and Bejís therefore qualifies.

Target region	Scar	Burned fraction	A whole region	A'	B scar area
				prevalence- matched	
Manavgat 2021	1	0.61	0.797	0.796	0.589
Bejís 2022	1	0.66	0.824	0.825	0.575
Muğla 2021	6	0.72	0.777	0.772	0.564
Muğla 2021	1	0.59	0.777	0.779	0.654
Muğla 2021	10	0.68	0.777	0.779	0.673
Muğla 2021	8	0.57	0.777	0.782	0.761
North Evia 2021	1	0.87	0.864	0.866	0.745
Montiferru 2021	1	0.58	0.720	0.717	0.622
Montiferru 2021	5	0.34	0.720	0.723	0.461
Mean			0.782	0.782	0.627

A minus A', the effect of prevalence alone, is **−0.000** [**−0.003**, **+0.002**]. A' minus B, the effect of replacing the region's negatives with fire-adjacent
1085 ones, is **+0.155** [**+0.093**, **+0.217**]. The evaluation-area effect is entirely the negative pool.

Appendix A.3. A(j). The transfer-gap decomposition, in full

Section 4.5 states the two figures this table carries — at most 34 % of the gap recovered in the directions that started below chance, and seven direc-
1090 tions with negative recovery. The table is kept here rather than in the body because Section 4.3 shows the within-region reference in its denominator is not matched to a transfer evaluation, and Section 4.4 shows its raw column is frame-dependent, so the fractions should be read as within-protocol quantities.

Table A5. Transfer-gap decomposition (four-AOI set, 12 direc-
1095 **tions).** Within = target's within-region thermal AUC; best adapted = the

better of z-score/CORAL; recovered fraction = (adapted - raw)/(within
 - raw), signed and unclipped, with paired bootstrap CI (1000 replicates).
 Montiferru directions are not part of this decomposition (per-pair absolute
 1100 decompositions exist without fraction CIs). The status column asks whether
 the *adapted* value clears chance and uses the 2-cell adapted intervals of Ta-
 ble B9. The adapted arms were not recomputed at the coarser blocking of
 Appendix A(c), which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.549 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

1105 *Appendix A.4. A(k). The thermal sign, stratified*

Section 4.4 reports that all five regions agree on a negative association between pre-fire surface temperature and burning, and that neither of the

two obvious confounders explains it. The per-region values are here. Signed AUC against **burned** within the 10 km collar; stratified columns pool within-
1110 stratum concordance over deciles of the named variable, weighting each stratum by its positive-negative pair count. Source `matched_frame_gap.csv`, recomputable by `paper/code/verify_matched_gap.py`.

Region	LST raw	LST within elevation	LST within NDVI	LST within distance	NDVI raw	NDVI within LST
Manavgat	0.386	0.402	0.440	0.505	0.621	0.542
Bejís	0.405	0.509	0.486	0.350	0.618	0.574
Muğla	0.332	0.363	0.404	0.367	0.652	0.547
Evia	0.286	0.327	0.279	0.319	0.663	0.380
Montiferru	0.376	0.392	0.368	0.485	0.582	0.405

1115 Three readings follow. The LST sign survives stratification within elevation in four of five regions and within greenness in all five, despite $r(\text{LST}, \text{NDVI})$ reaching -0.92 , so it is neither a lapse-rate proxy nor an inverse-greenness proxy. Within distance to the nearest burned cell it crosses 0.5 in Manavgat, at 0.505, and is attenuated to near-null in Montiferru, at 0.485
1120 from an unstratified 0.376 — the two regions where the collar leaves the least residual gradient — so in those two the agreed sign is largely a weaker version of the same spatial effect the collar was introduced to remove, entirely so in Manavgat. And the reciprocal adjustment runs one way only: NDVI reverses in two regions once temperature is held, while LST reverses in none
1125 once greenness is held.

Appendix A.5. A(l). The LST anomaly under the difference instrument

Section 4.4 reports that no reversal meets this paper’s strict criterion once evaluation frames are equalised, and that the LST anomaly nonetheless differs between regions on the weaker instrument Table B3’s note commits

the paper to. These are the four pairs that have opposite-sided point estimates and a difference interval excluding zero, all on `lst_anomaly_mean`. Signed AUC within the 10 km collar; the two regions are bootstrapped independently under the 10-cell spatial-block scheme and differenced, 1000 replicates, seed 42. Source `matched_frame_gap.csv`, recomputable by `paper/code/verify_matched_gap.py`.

Table A6. Between-region differences in the signed LST-anomaly association, 10 km collar.

Pair	AUC A	AUC B	Difference	95 % CI
Bejís vs Evia	0.392	0.584	−0.191	[−0.295, −0.083]
Evia vs Montiferru	0.584	0.400	+0.184	[+0.012, +0.321]
Manavgat vs Evia	0.462	0.584	−0.122	[−0.225, −0.020]
Bejís vs Muğla	0.392	0.507	−0.115	[−0.219, −0.002]

Nine of the ninety feature-by-pair differences clear zero, against about 4.5 expected at nominal 5 % under the null, and no multiplicity correction is applied; four of the nine are this feature and three of those involve Evia. The nine features are effectively two to three dimensions (Section 4.4), so the ninety comparisons are not independent either. This is reported as a weaker result than a reversal, not as a restored one.

Appendix A.6. $A(m)$. The same-geography event pair

Appendix A(m) states the result and its withdrawal; the design, the two structural asymmetries that have no analogue in the twenty-direction matrix, and the direction of the bias they impose are here.

This arm was designed to hold place fixed and vary only the fire, which would have separated regional concept shift from everything that differs be-

tween study areas. Muğla burned twice, in 2021 and again eleven months later, on the same grid and through the same processing chain, and the 2022 arm is the 2021 population with the 2021 scar removed: 41,730 rows / 2,911 burned for 2021 against 38,790 rows / 331 burned for 2022. **Positive-**
1155 **carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm,** and eleven is below the sixteen this design sets as its own floor, so the 2022 intervals are read as indicative exactly as the 20-cell row of Table 1 is. Full per-feature values are in Appendix B, Table B5.

On the frame as drawn, elevation reverses with bootstrap sup-
1160 **port.** In 2021 higher ground burned preferentially, at 0.611 [0.532, 0.690]; in 2022 lower ground did, at 0.296 [0.230, 0.355]. The intervals are disjoint and the difference is -0.317 [-0.414 , -0.220]: the same predictor, region and grid, and an association pointing the opposite way in two fires eleven months apart. The four absolute thermal channels change side as well.

Section 4.4 has already withdrawn it. The 2022 arm is one compact
1165 scar inside the whole Muğla box, so 93.2 % of its cells lie beyond 10 km of any burned cell against 55.3 % for the 2021 arm — the most extreme far field in the cohort. Under the same 10 km collar the 2021 figure barely moves, 0.611 to 0.606 and still supported, while the 2022 figure moves from 0.297
1170 to 0.565, onto the same side of 0.5 as 2021 with an interval covering chance. The arm shows the same artefact as the cross-region reversals rather than confirming them. That verdict covers **elevation**, and slope, which behaves the same way; the arm’s thermal channels carry no verdict at all, because the 2022 event’s own predictors are not exported and the reconstruction used
1175 here is valid only for the year-invariant ones (Appendix A(o)). We keep the

section because the structural properties below have no analogue elsewhere in the paper, and because this arm is what motivated the frame test.

The two arms are not disjoint samples: they share 38,789 of the 2022 arm’s 38,790 cells, and elevation, slope and land cover are identical to the
1180 digit across all 73,098 grid cells, so only NDVI and the six thermal channels carry new information between them. And the removal is the target’s own positive class, so in the 2022-to-2021 direction not one target positive is in the source training population while 38,789 of 38,819 target negatives are, and membership of the source training set alone separates the 2021 target’s
1185 classes at ROC-AUC 0.9996. **We therefore cannot bound what that asymmetry does to a transfer estimate between the two arms.** The known part of the bias runs the safe way: the removed cells are high, with a median of 563 m against the population’s lower centre, so removing them strips high negatives from the 2022 arm and makes the observed reversal
1190 smaller rather than larger.

Appendix A.7. A(n). The two interventions, in full

Appendix A(n) states both results; the four feature-set configurations, the per-region deltas and the pooled per-target shortfalls are here.

(a) Pooled multi-region training (Fig. A.1). At the point estimate,
1195 training on the pooled primary populations of the other four regions never beats the best single-source transfer for any target, with shortfalls of 0.02 to 0.22, and it stays 0.28 to 0.50 AUC below the within-region ceiling; for two targets it falls below the pairwise mean and below chance, though only Bejís is below chance with interval support, at 0.417 [0.369, 0.467]. Aggregation
1200 does not recover what single-source transfer loses.

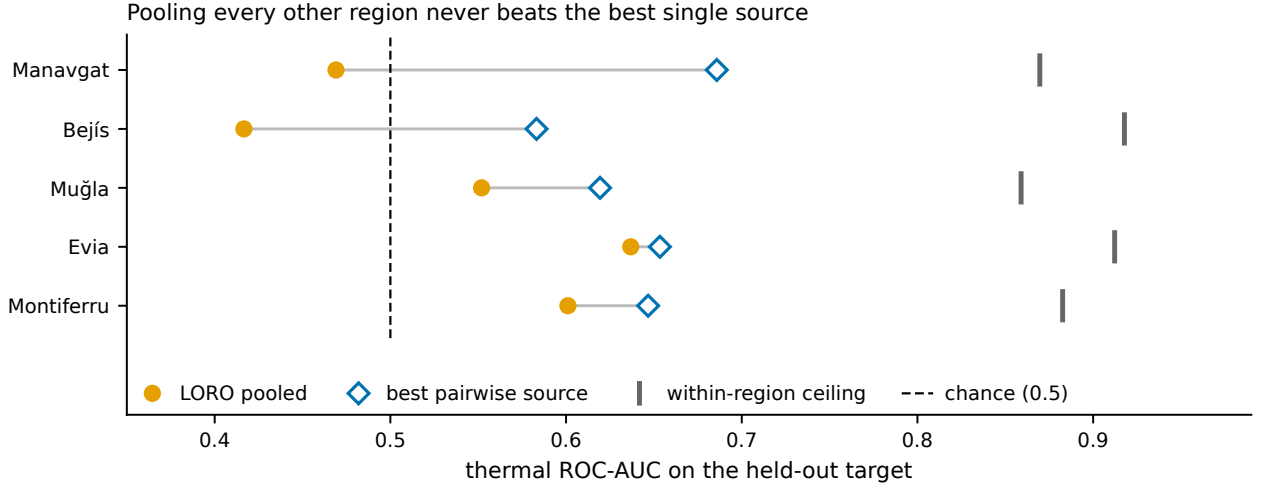


Figure A.1: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

(b) **Removing the direction-reversing features.** The two predictors whose signed association reverses between regions **with bootstrap support on the frames as drawn** are `elevation_mean` and `lst_anomaly_mean`. Section 4.4 withdraws that support under an equalised frame, so this selection rule is frame-dependent and the arm below measures what the removal costs under the original protocol, not a consequence of an established reversal. The per-region signed associations and the three pair-level reversals that meet that criterion are in Appendix B, Tables B2 and B3. Retraining

without them costs **-0.081** of mean within-region AUC, supported in every
1210 region (per-region deltas -0.060 , -0.130 , -0.074 , -0.063 , -0.079 , every in-
interval entirely below zero), and changes mean transfer by **+0.014** [**-0.017**,
+0.045], which spans zero.

Two qualifications belong with those numbers rather than after them.
First, the debit is not the thermal block's. Dropping elevation alone accounts
1215 for -0.061 of it, from 0.888 to 0.827 . Dropping the LST anomaly alone
accounts for -0.013 , from 0.888 to 0.875 . Roughly three quarters of the
cost is therefore the removal of a **baseline** terrain variable, not of a thermal
channel. Second, both figures are post-selection. The two predictors were
chosen because they reverse, using the same data on which the -0.081 and
1220 the $+0.014$ are then estimated. No correction for that selection is applied.

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.888	0.541
drop <code>elevation_mean</code>	0.827	0.546
drop <code>lst_anomaly_mean</code>	0.875	0.544
drop both	0.807	0.556

Fig. A.2 shows the four configurations together. What this measures is
a local cost with no compensating transfer gain. It does not measure an
exchange, because the transfer side is a null on both arms. The thermal
1225 block's own contribution to transfer is $+0.004$, with an interval spanning
zero (Section 4.5). Removing the reversing predictors returns $+0.014$, with
an interval spanning zero. Two nulls on the portability axis are not a price
paid.

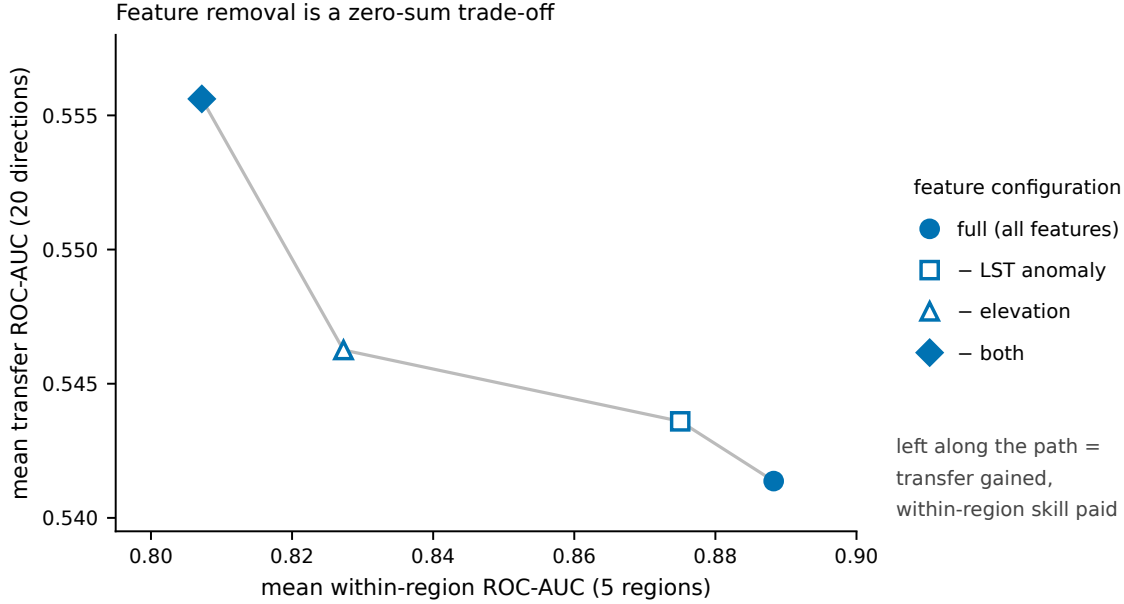


Figure A.2: **Removing the direction-reversing features costs locally and returns nothing measurable on transfer.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically ($0.541 \rightarrow 0.556$) while within-region skill falls monotonically ($0.888 \rightarrow 0.807$). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The within-region cost is 0.081 AUC and is supported in every region. The change in transfer is +0.014 AUC on an interval that spans zero, so no exchange between the two is demonstrated: a local cost is measured and no compensating transfer gain is.

Appendix A.8. A(o). The thermal sign and the anomaly difference, in full

1230 Section 4.4 states both results; the per-region values, the reciprocal stratification and the difference-instrument argument are here.

A weaker instrument does support the anomaly result. Table B3's note commits this paper to a difference interval on the pair as the sharper test, and it is applied here. Bootstrapping the two regions independently under the collar and differencing, four pairs have opposite-sided point estimates **and** a difference interval excluding zero, all on `lst_anomaly_mean` and three of the four involving Evia — the largest being Bejís against Evia at -0.191 $[-0.295, -0.083]$ (Appendix A(1), Table A6). The anomaly is the one channel the frame-and-terrain mechanism of this section cannot explain, being decorrelated from elevation. Three caveats keep it weak and all three are stated rather than buried: nine of ninety feature-by-pair differences clear zero against about 4.5 expected under the null with no multiplicity correction; the nine features are effectively two to three dimensions, as the collinearity paragraph below shows; and Evia's own interval-support status turns on 0.003.

1240 **The honest statement is that no reversal meets this paper's strict criterion under the collar, and that the LST anomaly differs between regions on the weaker difference instrument.** We state it that way because the alternative would be to apply a looser standard to the arm that supersedes Table B3 than to Table B3 itself. Elevation under the collar is above 0.5 in all five regions but individually supported in only two, Muğla at 0.606 $[0.525, 0.685]$ and Evia at 0.648 $[0.550, 0.740]$.

1250

The sign the five regions agree on is not the one the dryness framing predicts. For LST the common direction is *below* 0.5 in every

region, at 0.386, 0.405, 0.332, 0.286 and 0.376: a hotter pre-fire surface is
1255 associated with **less** burning, and the same holds for TVDI. It is not a lapse-
rate artefact and it is not greenness acting through fuel load. On mutual
adjustment the surviving channel is LST, not NDVI: holding NDVI, LST
never reverses in any region, while holding LST, NDVI's own association
reverses in Evia and Montiferru. Greenness is therefore not the mechanism.
1260 Two caveats bound what the sign is: in Manavgat and Montiferru it is a
residual spatial gradient that disappears when distance to the nearest burned
cell is stratified within the collar, and interval support is not uniform, LST
being supported in four of five regions and TVDI in three, so “all five agree” is
a statement about point estimates. What we can defend is that the absolute
1265 thermal channels behave here as **static land-surface descriptors** rather
than as a dryness index, and that the two internally differenced channels
carry no consistent cross-region direction at all. Compositing depth was
not tested and remains an open alternative. Per-region stratifications are
in Appendix A(k), and Section 5.2 states the consequence for the moisture-
1270 stress motivation of Section 1.2.

**The same-geography arm of Appendix A(m) is the most extreme
case in the cohort, and it does not survive either.** A fixed study area
does not mean a fixed evaluation frame: its 2022 arm has **93.2 % of cells
beyond 10 km of any burned cell, median 43.6 km**, a larger far field
1275 than any cross-region arm. Applying the same collar (`mugla_two_event_
collar.csv`; the full-frame values reproduce Table B5 on the year-invariant
channels only, for the reason given below):

What this arm can and cannot evaluate. The 2022 event's own

step8a export is not in this tree. The released script (`code/verify_mugla_`
 1280 `collar.py`) reconstructs the arm by taking the **2021** predictor file and sub-
 stituting the 2022 burned mask, justified on the ground that elevation, slope
 and land cover are identical between the arms. That justification holds for
 those channels and **not for the state-dependent ones**, which the script
 nonetheless computes. Checked against the frozen step9g export, which uses
 1285 the 2022 arm’s own predictors, the reconstruction agrees on the year-invariant
 channels — elevation 0.297 against 0.296, slope 0.559 against 0.558 — and
 diverges on every seasonal one, by +0.136 on `lst_anomaly_mean`, +0.092
 on `tvdi_difference_mean`, −0.078 on `ndvi_mean` and −0.025 on `current_`
`tvdi_mean`. **So only elevation and slope can be given a collar verdict**
 1290 **for this arm**, and the thermal channels cannot until the 2022 arm’s own
 predictors are exported.

Feature	2021, full	2022, full	full	2021, collar	2022, collar	collar
			ver-			ver-
			dict			dict
elevation	0.296 [0.159, 0.692]	0.297 [0.229, 0.363]	supported	0.661 [0.525, 0.685]	0.565 [0.450, 0.677]	none
			re-			
			ver-			
			sal			
slope	0.558 [0.584, 0.689]	0.559 [0.463, 0.643]	none	0.635 [0.578, 0.692]	0.457 [0.356, 0.548]	none
the —	—	—	—	—	—	not
five						evalu-
state-						able
dependent						here
chan-						
nels						

On the two channels this arm can evaluate, the collar removes
the reversal. Elevation’s 2022 arm crosses to the same side of 0.5 with an
 1295 interval covering chance, while the 2021 arm is nearly frame-invariant, which

isolates the effect to the 2022 arm’s far field rather than to the collar. Slope reverses on neither frame.

Two corrections to earlier statements belong here. First, an earlier version generalised the elevation row to “no bootstrap-supported sign reversal survives anywhere in this paper once evaluation frames are equalised”. **That**
1300 **generalisation is withdrawn:** it was computed for one feature and the other five could not have supported it either way. Second, a subsequent version reported `current_lst_mean` and `current_tvdi_mean` as supported reversals appearing under the collar. **Those are withdrawn too** — they
1305 are the 2021 pre-fire surface scored against the 2022 scar, not the 2022 fire’s own state. The defensible statement is narrower than either: **on the channels this design can evaluate, the collar removes this arm’s reversal; on the rest the arm is silent.** Whatever it shows rests on eleven positive-carrying blocks against this design’s own floor of sixteen, so no verdict from
1310 it is strong.

Two points bound how many independent reversals could have been counted in the first place. The channels whose reversal vanishes are the ones partly measuring terrain: current LST correlates with elevation at -0.695 to -0.125 across the regions and TVDI at -0.722 to -0.298 , so their proxy strength
1315 varies fivefold and a frame that shifts the elevation distribution shifts them with it. And within the collar `fused_lst_mean` correlates with `current_lst_mean` at 0.99 to 1.00 , `downscaled_lst_mean` at 0.97 to 0.99 and `current_tvdi_mean` at 0.87 to 0.98 , so the four absolute channels are close to one axis and the two differenced channels correlate at 0.64 to 0.94 : the “five of nine
1320 directions reverse” count of Appendix A(s) counts features, not independent

quantities, and **in effective dimensions it is closer to two.**

Two scope statements belong with the table. The full-frame values here are not numerically the published ones — $\rho = +0.86$ over fourteen directions against the $+0.84$ over sixteen in Section 4.6 — because the support test
1325 is itself bootstrap-dependent and at this setting three region pairs carry no jointly supported feature rather than two; the argument rests on before-and-after under identical settings, not on the published figure. And **only these diagnostics were recomputed on the collar**: the other eighteen failed on the frames as drawn and were not rerun, so their correlations against the
1330 equalised transfer vector are unknown rather than shown to be null.

The thermal block’s paired contribution to transfer, with its interval. The two matrices were differenced direction by direction. The mean is **+0.004**, but the mean is not the informative statistic here. The individual paired contributions span **−0.148 to +0.133**, with **twelve positive**
1335 **and eight negative** (`baseline_vs_thermal_transfer.csv`). The spread is thirty times the mean, and the sign is not a property of the block but of the pair it is asked to cross: the same six predictors that add $+0.133$ in one direction subtract 0.148 in another. A mean near zero here records cancellation, not consistent absence of effect. The directions are not independent,
1340 because each region appears in eight of the twenty. The interval therefore depends on what is treated as the resampling unit. All four units the design permits give the same answer:

Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive	20	$[-0.027, +0.034]$
Unordered pairs, cluster bootstrap	10	$[-0.028, +0.036]$
Unordered pairs, t on pair means	10	$[-0.034, +0.042]$
Regions, leave-one-out jackknife	5	$[-0.037, +0.046]$

Every interval spans zero, and the point estimate is a small fraction of each width. The per-region jackknife shows how little the mean is anchored. Holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives +0.0148, +0.0021, +0.0065, **-0.0081** and +0.0060. **Dropping Evia alone reverses the sign of the headline.** None of these four units propagates within-direction sampling variability. They resample between directions only.

In precision terms the transfer is worse than the ROC figures suggest. ROC-AUC is the metric used throughout this paper, for comparability with the literature, but a susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity. Across the twenty directions the thermal model’s PR-AUC averages **0.156 against a no-skill baseline of 0.136.** The mean of the twenty per-direction lifts is 1.16; the ratio of the two means just quoted is 1.146. **Six of the twenty fall below their own no-skill baseline at the point estimate, and five of those six have intervals entirely below it.** The exception is Bejís to Manavgat, at 0.034 $[0.029, 0.042]$ against a baseline of 0.038, whose interval covers the baseline and which is therefore a point-estimate case only. Only one direction, Evia to Manavgat, exceeds twice its baseline, at 0.094 against 0.038. A transferred model therefore ranks burned cells about a sixth better than

random on average, and worse than random in five directions with interval support and a sixth at the point. This is a sharper statement than the ROC means support and it should be the one a practitioner reads.

The within-region increment survives the same correction, and is reported here because Section 4.2 establishes it on the frames this section calls incomparable. Running the baseline arm on the collar as well (`collar_increment_and_cosine.csv`, `paper/code/verify_collar_increment.py`), the thermal increment at 5 km blocking is +0.041, +0.030, +0.091, +0.134 and +0.090 across the five regions, **positive in all five**, with a mean of +0.077 against +0.086 on the frame as drawn. It does erode as the frame tightens further: at a 5 km collar the mean halves to +0.041, still positive in all five. These are point estimates; the intervals in Table 1 are computed on the frame as drawn. The paper’s one surviving predictor-level positive claim therefore holds on the frame it argues is the correct one.

The five areas of interest are not comparable frames. Each region is a rectangle drawn around a fire, and the rectangles differ by an order of magnitude in how much unburnt far field they enclose.

The share of modelled cells lying beyond 10 km of any burned cell is 60.1 % in Manavgat, 63.1 % in Bejís, 55.3 % in Muğla, 43.7 % in Evia and **2.1 %** in Montiferru, with median distances of 13.4, 13.5, 11.3, 8.0 and 2.7 km (Appendix B, Table B6). Montiferru’s frame is fire-scale; Manavgat’s and Bejís’s are roughly three-fifths far field. The far field is not a neutral addition. In Manavgat the median elevation of modelled cells rises from 472 m within 5 km of the fire to 955 m at 10 to 20 km and 1,273 m at 20 to 50 km, against 512 m for the burned cells themselves, in Taurus terrain that no

plausible spread model would place at risk.

Appendix A.9. A(s). The contrast pair, in full

1390 The clearest single view needs no ranking at all (Fig. A.3). Manavgat and Muğla lie in the same country and fire year, 306 km apart by centroid, and their burned cells occupy the most similar environmental envelope of any pair in the matrix; Bejís and Montiferru occupy the least similar.

Per-quantity values for both pairs, on both frames, are in Appendix B,
1395 Table B10.

Two readings of this pair do not survive Section 4.4 and are not offered. On the frames as drawn five of nine feature-response directions point opposite ways, elevation among them; under the collar that figure moves to 0.561 against 0.606, on the same side of 0.5. And transfer is below chance in both
1400 directions as drawn but above chance under the collar. **What survives is the ordinal contrast:** the most similar pair is among the weakest in the matrix and the least similar among the stronger, while neither is the extreme — the weakest direction is Manavgat to Bejís at 0.417 and the strongest Muğla to Evia at 0.727. The claim this supports is that high
1405 envelope overlap does not buy transfer, not that it produces anti-prediction. The AoA shares are full-frame quantities.

Bejís and Montiferru sit at the opposite extreme. Their burned envelopes barely overlap, and they carry the most dissimilar values on every overlap measure. They transfer above chance in both directions. That half of the
1410 contrast rests on point estimates: neither direction carries a verdict at 5 km blocking, and the marginal applicability audit was never produced for

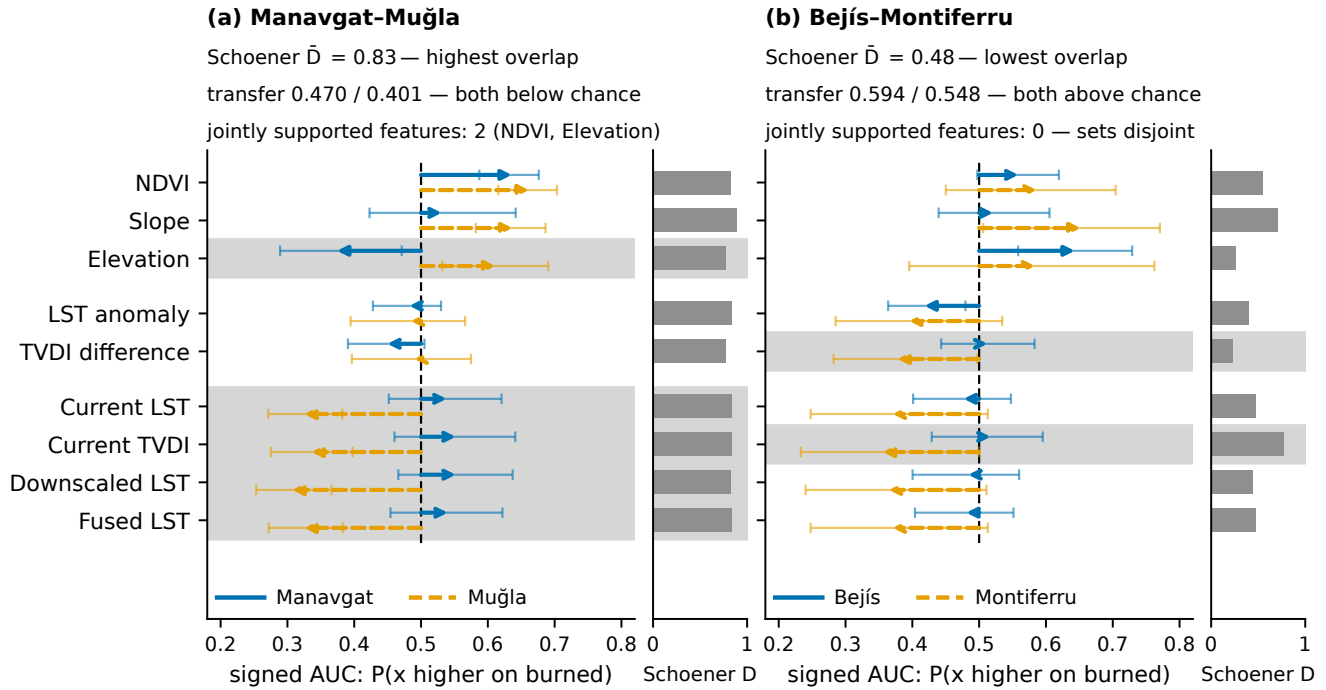


Figure A.3: **Burned-niche overlap does not determine transfer: the contrast pair.** Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions on the frames as drawn (0.470 and 0.401); under the equalised frame of Section 4.4 it is 0.551 and 0.510, above chance but still among the weakest in the matrix. **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 as drawn, 0.669 equalised, and 0.548 against 0.624). High niche overlap is therefore not sufficient for transfer, which is an ordinal claim on either frame. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals are wide. Pair (b) therefore drops out of the supported-index sample entirely. The index is undefined exactly on the pair that most challenges the overlap explanation, which is a limitation of the index and not evidence for it.

Montiferru. The claim is one of *sufficiency*. Similarity does not guarantee transfer, and dissimilarity does not preclude it. The claim rests on two coexisting counterexamples, so it does not depend on the number of pairs available.

Appendix A.10. $A(v)$. The sensitivity arms, summarised

Eight design choices were varied with everything else held fixed. They are the Evia AOI and its prevalence, the CORAL regularisation constant, the blocking scale, the closure date of the predictor window, the quality screening of the coarse thermal input, the contrast between the normalised and the absolute dryness channels, the removal of the coordinate-informed channels, and the capacity of the classifier. None changes a conclusion above. Two bound how the results should be read, so they are carried into the main text here.

The secondary population. Section 3.5 defines a secondary population of all valid cells, including cropland. The frozen export carries it for the within-region arm in Manavgat and Bejís only, so it is a two-region sensitivity rather than a cohort-wide one, and no transfer quantity is defined on it. Across the two regions and the three block sizes the thermal increment runs +0.044 to +0.061 with every bootstrap interval above zero, against +0.046 to +0.067 on the primary population, and the paired difference between populations runs -0.008 to $+0.011$ with no consistent sign (`experiments/<region>/step10/within_robustness.json, by_population`). The increment is therefore not an artefact of excluding cropland, which is what this arm was run to test; it says nothing about the transfer results, where the primary population is the only one available.

Coarsening the blocks from 1 km to 5 km moves the **paired thermal-minus-baseline delta** verdicts from ten positive, seven negative and three uncertain to six, four and ten. The above-chance verdicts on the thermal arm itself, a different quantity, move from twelve, six and two to nine, four and seven. Support is removed from seven verdicts and added to none. The point estimates are unchanged, but that is an identity rather than a result. The blocking scale is the resampling unit, and it cannot move an estimate computed once over all target cells.

Manavgat’s whole downstream chain was then rebuilt from a quality-screened MODIS input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. This closes the one processing-artefact candidate for that region’s behaviour. Appendix A reports all eight arms, including one that tests a claim of Section 1.2 and does not uphold it.

Appendix A.11. $A(t)$. Distance within a region

One further arm measures how skill decays with distance inside a single region. A model is fitted on one half of a region and applied to the other, and target cells are binned by their distance from the training cells (`distance_curve.md`, `paper/code/distance_curve.py`). Means are unweighted over bins, whose positive counts range from 1 to 2,564, and the bins carry no intervals.

Separation from training cells	Bins	Mean target AUC
0 to 5 km	18	0.692
5 to 10 km	16	0.519
10 to 20 km	11	0.499
20 to 40 km	6	0.445
40 to 80 km	3	0.541
80 to 160 km	1	0.421
cross-region, 306 to 2,802 km	20	0.541

By 10 to 20 km inside a single region the model is already at
1460 **chance.** That bounds how far a susceptibility surface of this kind can be
carried from the cells it was fitted on, and it is consistent with Section 4.3,
where withholding a scar and replacing the model with a foreign one cost
nothing distinguishable.

It does not reframe the paper’s negative result, and a reading
1465 **that it does was considered and rejected.** That reading argued that
because the twenty cross-region directions average 0.541, at or above the
within-region plateau, they sit on the continuation of the curve and the fail-
ure needs no regional mechanism — a rule fixed in advance in `positive_`
`control.md`. Two objections defeat it, and `scar_control.md` records the re-
1470 framing as withdrawn in full. **The rule cannot fail:** once the curve reaches
the chance floor, any cross-region mean near 0.5 lies on its continuation by
construction, so the comparison could not have come out otherwise, and a
test that cannot fail is not evidence. And **extrapolating an uninforma-**
tive model does not produce a reliably reversed ranking: Manavgat
1475 to Bejís is below chance with interval support on the frame as drawn and at
the 10 km collar, which is not what a model that has merely run out of skill

returns.

Four limits bound even the descriptive reading. The two distance ranges do not overlap — within-region separations span 2 to 86 km, cross-region
1480 separations start at 306 km — so any comparison across the gap extrapolates the curve. The far bins are thin, six at 20 to 40 km and one beyond 80 km, so their means should not be read closely and the apparent rise at 40 to 80 km is not evidence of anything. The near bins carry exactly the autocorrelation blocked validation exists to remove, so **0.692 is an upper bound on near-**
1485 **field skill rather than an estimate of it.** And this design separates distance from crossing a study-area boundary, but not from the land cover, terrain and fire history that covary with it.

The arm therefore contributes a length scale for the within-region decay, not an attribution. The unit that fails to transfer is not established by this
1490 design, and Section 4.3 says so directly.

Appendix A.12. $A(u)$. What target labels cost

Everything above measures a failure; this prices it. Label-free alignment does not close the residual gap, so the missing resource is information about the target that alignment cannot synthesise, and the direct way to supply
1495 it is target labels. A frozen labelled-budget diagnostic answers how many, for three regions across all six ordered directions, using one 10-cell (~ 5 km) spatial block as the unit of labelling effort and reading recovery against a matched target-only ceiling of 0.777 to 0.824.

Thirty-two labelled blocks recover **85 to 89 % of that ceiling in three**
1500 **of the six directions**, two of which started below chance; 51 to 57 % in two more, the directions where the conditional gap is widest; and 30 % in the

sixth. That budget is 7 to 20 % of the target’s natural-vegetation population, which at the 2,700 to 3,000 labelled cells those blocks carry (S1.3) and this grid’s effective cell area of 0.199 to 0.208 km² (Appendix C) is roughly **540**
1505 **to 620 km²**, so it is a real answer and not a cheap one. Two properties of the measurement bound it and are stated in the supplement rather than buried: at the top budget the selection pool is nearly exhausted, so the narrow upper-budget intervals reflect saturation rather than precision, and the labelled blocks are drawn from the event being predicted, which is not a
1510 resource available before that event burns. The protocol, the full curve and the limits are in Supplement S1.

Two limits belong with the number. At small budgets the same intervention damages the direction that already transfers best without any labels. And six directions in three regions cannot support a general label budget, so
1515 none is offered.

Row D is a mean over four foreign models, and the spread behind it is large. Decomposing it gives **32** scar-by-source combinations running from 0.374 to 0.724, with **ten below chance** and a mean spread of 0.171 across the four sources for a given scar (Appendix A, Table A3). The
1520 pooled 0.555 is therefore not a statement that any foreign model does as well as a same-region one, but that the *average* one does, and the variation it averages over is exactly the pair-specific instability that the rest of Section 4 is about.

One confound must be stated. Holding out a region’s only large scar
1525 also removes most of its positives: the source model retains 11 positives in Evia, 88 in Manavgat and 97 in Montiferru, against about 2,000 in Muğla,

which has four separate scars. Those four non-Muğla arms average 0.525 and the four Muğla arms 0.579. Three of the four are genuinely starved — the fourth, Montiferru’s second component, retains 472 source positives — so row C mixes “the fire was withheld” with “almost all the positives were withheld”. Restricted to Muğla, C is 0.579 and D is 0.597, and the C-to-D comparison still shows nothing. **With eight scars, three of them starved, this design cannot establish a fire-specific residual, only bound it at about 0.18.**

What this establishes is bounded rather than positive. Rows C and D both sit close to chance, so the comparison between them has little dynamic range, and neither the fire’s identity nor the region boundary is shown to cost anything here. What is measurable is the change of evaluation geometry between rows A and B. Nor does it establish the fire event as the unit, because the held-out patch is defined by the labels, so its identity cannot be separated from its location. The within-region half-split, which does see half of the target scar, returns 0.574 against leave-one-scar-out’s 0.552, with overlapping intervals.

The result is robust to how the held-out patch is defined. Sweeping the two parameters that define it — minimum burned-component size over 25, 50, 100 and 200 cells, and connectivity over the 4- and 8-neighbourhoods — moves the row-C mean between **0.545 and 0.566** across all eight settings, against the 0.552 of the reported configuration, which the sweep reproduces to 0.001 (`scar_definition_sweep.csv`, `paper/code/scar_definition_sweep.py`). The number of qualifying scars falls from eleven to seven as the size threshold rises, which is what moves the mean; connectivity changes it by at

most 0.001 at any threshold. The result is also flat across buffers of 2, 5 and 10 km.

A weaker instrument does support the anomaly result. Table B3's
1555 note commits this paper to a difference interval on the pair as the sharper
test. Applied under the collar, four pairs have opposite-sided point estimates
and a difference interval excluding zero, all on `lst_anomaly_mean` and three
of the four involving Evia (Appendix A(1), Table A6). Three caveats keep it
weak: nine of ninety feature-by-pair differences clear zero against about 4.5
1560 expected under the null with no multiplicity correction, the nine features are
effectively two to three dimensions, and Evia's own interval-support status
turns on 0.003. **The honest statement is that no reversal meets this
paper's strict criterion under the collar, and that the LST anomaly
differs between regions on the weaker difference instrument.** Eleva-
1565 tion under the collar is above 0.5 in all five regions but individually supported
in only two.

**The sign the five regions agree on is not the one the dryness
framing predicts.** In every region a hotter pre-fire surface is associated
with **less** burning, and the same holds for TVDI. It is neither a lapse-rate
1570 artefact nor greenness acting through fuel load: on mutual adjustment the
surviving channel is LST, not NDVI. Two caveats bound it — in two re-
gions it is a residual spatial gradient that disappears when distance to the
nearest burned cell is stratified inside the collar, and interval support is not
uniform, so “all five agree” is a statement about point estimates. What
1575 we can defend is that the absolute thermal channels behave here as **static
land-surface descriptors** rather than as a dryness index, and that the two

internally differenced channels carry no consistent cross-region direction at all. Compositing depth was not tested and remains an open alternative. Per-region values are in Appendix A(k); Section 5.2 states the consequence for the moisture-stress motivation of Section 1.2.

Appendix A.13. A(w). The frame test, elaborated

Section 4.4 states these results; the paragraphs it condensed are here.

The five areas of interest are not comparable frames. Each is a rectangle drawn around a fire, and they differ by an order of magnitude in how much unburnt far field they enclose: the share of modelled cells beyond 10 km of any burned cell runs from **2.1 %** in Montiferru to **63.1 %** in Bejís (Appendix B, Table B6). That far field is not a neutral addition — in Manavgat the median elevation of modelled cells rises from 472 m within 5 km of the fire to 1,273 m at 20 to 50 km, against 512 m for the burned cells themselves.

Under an equalised frame the sign reversals of Appendix A(s) do not survive. Restricting every region to cells within 10 km of any burned cell removes only far-field negatives; every burned cell is at distance zero and is retained at any radius, so the protection against choosing a flattering radius is the sweep below, not the retention of positives. On the equalised frame all five regions agree in sign on elevation, on LST and on TVDI, and both bootstrap-supported elevation reversals of Table B3 disappear. Under this paper’s own criterion from Section 3.10 — each region’s own 10-cell block-bootstrap interval must exclude 0.5 — **no between-region reversal remains bootstrap-supported.** The qualifier matters: this arm covers the five regions, and the same-geography two-fire arm is treated separately

in Appendix A(o), where only the year-invariant channels can be given a verdict at all. Two features straddle 0.5 at the point estimate and neither is supported: `lst_anomaly_mean` at 0.392 [0.324, 0.460] in Bejís against 0.584
1605 [0.497, 0.669] in Evia, where Evia’s interval includes 0.5 by 0.003; and `tvdi_difference_mean` at 0.509 [0.417, 0.597] in Muğla against 0.384 [0.288, 0.504] in Montiferru, where no region’s interval excludes 0.5. These are the two internally differenced channels. An earlier draft treated the first as uniquely informative because it carries no lapse-rate signal; that does not hold, since
1610 the second is equally decorrelated from elevation and behaves the same way.

A weaker instrument does support the anomaly result. Table B3’s note commits this paper to a difference interval on the pair as the sharper test; applied under the collar, four pairs have opposite-sided point estimates **and** a difference interval excluding zero, all on `lst_anomaly_mean`
1615 and three of the four involving Evia. Three caveats keep it weak: nine of ninety feature-by-pair differences clear zero against about 4.5 expected under the null with no multiplicity correction, the nine features are effectively two to three dimensions, and Evia’s support status turns on 0.003. **The honest statement is that no reversal meets this paper’s strict criterion**
1620 **under the collar, and that the LST anomaly differs between regions on the weaker instrument** (Appendix A(l)).

The sign the five regions now agree on is not the one the dryness framing predicts. In every region a hotter pre-fire surface is associated with **less** burning, and the same holds for TVDI. It is neither a lapse-rate
1625 artefact nor greenness acting through fuel load: on mutual adjustment the surviving channel is LST, not NDVI. Two caveats bound it — in two regions

it is a residual spatial gradient that disappears when distance to the nearest burned cell is stratified inside the collar, and interval support is not uniform, so “all five agree” is a statement about point estimates. The absolute thermal channels therefore behave here as **static land-surface descriptors** rather than as a dryness index, and the two internally differenced channels carry no consistent cross-region direction at all. Compositing depth was not tested and remains an open alternative. Per-region values are in Appendix A(k); Section 5.2 states the consequence for the motivation of Section 1.2.

The same test determines what the diagnostics of Section 4.6 can establish. The only two candidates there with intervals excluding zero measure agreement in the sign of each predictor’s association between source and target, counted over features interval-supported in both. Those are built out of exactly the signed AUCs this section has shown to be frame artefacts, and in Section 4.6 they are correlated against transfer measured on the same unequal frames. Recomputing both sides under the 10 km collar gives Table 3 of the body (Section 4.4), which is not repeated here.

The two fail differently and both fail. Once frames are equalised every region pair agrees in sign on every jointly supported feature, so the agreement fraction has no variance left. The supported cosine keeps a trace of variance — unanimous signs fix directions but not magnitudes — and simply stops tracking transfer. This is not a marginal shift: features supported in both regions *rise* from 1.20 to 3.40 per direction, so the diagnostics are better determined and unanimous. The disagreements they were reading were the far fields.

Two scope statements belong with Table 3 and are given in Appendix

A(o): the full-frame values here are not numerically the published ones, because the support test is itself bootstrap-dependent; and only these diagnostics were recomputed on the collar, so the other eighteen are unknown
1655 against the equalised transfer vector rather than shown to be null.

This settles what Contribution 3 can claim. It is not that conditional similarity orders transfer where marginal similarity fails; it is that **no diagnostic tested here has been shown to order transfer**. The eighteen marginal, niche and regime candidates do not order it on the frames as drawn,
1660 and the two conditional variants that do stop doing so once the frames are comparable, because what they were reading is how the study rectangles were drawn. The practitioner’s position is therefore worse than the as-drawn analysis suggests, not better: there is no screen, and the apparent exception is an artefact. Section 4.6 reports its numbers as computed under the
1665 pre-registered protocol, which is what a reader following that protocol would obtain.

The same-geography arm of Appendix A(m) does not survive either, and it is the most extreme case in the cohort. A fixed study area is not a fixed evaluation frame: its 2022 arm has 93.2 % of cells beyond
1670 10 km of any burned cell against 55.3 % for 2021. Under the same collar the 2021 elevation figure barely moves while the 2022 figure crosses to the same side of 0.5 with an interval covering chance, so the elevation reversal does not survive equalisation. **This does not generalise across features:** on the same arm two absolute thermal channels become supported reversals under
1675 the collar that were not reversals as drawn, so equalisation moves this arm’s reversal rather than removing it. The seven-feature table is in Appendix

A(o); values in Appendix A(m).

The within-region increment survives the same correction, remaining positive in all five regions on the equalised frame with a mean of
1680 +0.077 against +0.086 as drawn, and halving but staying positive in all five at a 5 km collar. The paper’s one surviving predictor-level positive claim therefore holds on the frame it argues is the correct one. Per-region values are in Appendix A(o).

The data-provenance correction. Muğla’s 500 m modelling dataset
1685 at the canonical path had been overwritten by a quality-screening rebuild after the frozen tables were computed, so this section’s scripts had read a version of that one region differing from Table 1 and B9 in `downscaled_1st_mean` and `fused_1st_mean`. The two files agree on every other column; the proof that the difference is the data and not the fitting is that both
1690 give a within-region baseline of 0.6980 at 10-cell blocking while the thermal arm gives 0.7834 against the frozen 0.7773. Two regions’ files carry the rebuild’s timestamp, Manavgat’s and Muğla’s, so both were replaced; the distinction that matters is whether the replacement agrees with what was published. Manavgat’s does, reproducing its Table 1 row to the printed pre-
1695 cision at 0.748 and 0.797, as does the rebuild’s own reported increment interval. Muğla’s does not. Bejís, Evia and Montiferru were not touched. Every arm of Section 4.4 was therefore re-run with Muğla read from the surviving frozen copy and the other four regions as they stand, which is the configuration Table 1 and B9 were computed under. The correction moves forty of the
1700 hundred per-direction transfer values by up to 0.022, the largest being Muğla to Montiferru at $0.509 \rightarrow 0.531$. It leaves every headline quantity of Table 4

unchanged to within 0.0012, the largest being the full-to-10 km paired delta at +0.0021 against +0.0033: the equalised mean 0.6163 against 0.6166, its paired delta +0.0231 against +0.0234, and the above-chance counts identical in four rows of five, the 5 km row moving from 18 to 19 as one direction crosses 0.5. The signed-AUC results are unaffected in substance — only Muğla’s two channels move, by at most 0.008, and the same two features straddle 0.5 on each frame. Sources `aoi_frame_transfer_frozen_mugla.csv`, `aoi_frame_auc_frozen_mugla.csv`, `frozen_mugla_recheck.json`; code `code/frozen_mugla_verify_aoi_transfer.py`.

The transfer matrix moves as well. Restricting source and target to the same collar gives Table 4 of the body (Section 4.4), which is not repeated here.

The reference arm reproduces the frozen matrix, at 0.541 against Table B9’s 0.541 and 14 of 20 exactly, so this is measuring the same quantity. **The baseline control moves with it and must be restated on this frame:** the static baseline transfers at 0.593 against the thermal model’s 0.616, a paired difference of +0.023 rather than the +0.004 of the frame as drawn. The control still holds in kind — the static predictor class is not the portable one either — but the gap between them is about six times larger once frames are comparable, and Sections 1, 5 and 6 quote only the as-drawn pair. Table 4’s above- and below-chance counts are point counts. Under the same 10-cell block bootstrap used for Table B9, at 1000 replicates, the full frame gives nine directions above chance and four below with interval support, and the collar frame fifteen above and one below, so the headline movement is six to one at the point estimate and **four to one with interval support** (`aoi_`

`frame_transfer.csv`, which carries the per-direction bounds). The largest movers are Bejís to Evia, 0.383 to 0.602, and Bejís to Manavgat, 0.440 to 0.601.

1730 **What this settles, and what it leaves standing.** Five quantities re-
 1735 ported in the sections that follow are properties of the frames rather than of
 the predictor-burning relationship, and are identified as such where they ap-
 pear: the count of six anti-predictive directions, which becomes one; the sign
 reversal of elevation, LST and TVDI as a mechanism; the sign-agreement di-
 1740 agnostic of Section 4.6; the same-geography arm of Appendix A(m); and the
 paired thermal contribution, which is +0.004 as drawn and +0.023 equalised.
 What the correction leaves standing is the central negative result, and its size
 must be stated on a matched comparison. Setting the equalised transfer mean
 of 0.616 against a within-region reference of about 0.87 would compare a 10
 1740 km-collar number with a full-rectangle one at 1 km blocking, which is the
 most generous reference in the paper and the very figure Section 4.3 argues
 is an upper bound. Recomputing the within-region reference on the same
 frame and at the 5 km blocking this design defends (`matched_frame_gap.`
`csv`, `paper/code/verify_matched_gap.py`):

Frame	Within-region (5 km blocking)	Mean transfer	Gap
1745 full rectangle	0.798	0.540	0.258
10 km collar	0.772	0.616	0.155
5 km collar	0.737	0.608	0.129

Paired by target region, the collar shortfall is **+0.155** [**+0.094**, **+0.217**]
 (Student t over the five target regions; per-region +0.086 Montiferru, +0.127
 Manavgat, +0.161 Muğla, +0.196 Bejís, +0.206 Evia), which reproduces the

point estimate above and excludes zero. The shortfall therefore survives
1750 on every matched row, but it is **0.155 at the collar, not the 0.25 the
unmatched comparison implies**, and it shrinks as the frame approaches
the fire — which is where a susceptibility surface is actually used.

This is Section 4.3’s effect acting between regions rather than within one,
on frames whose fire-adjacent share ranges from 37 % to 98 %. Appendix
1755 C.5(x) records the frame as a limitation of this cohort rather than of the
method.

Appendix A.14. $A(x)$. The transfer matrix and adaptation, elaborated

Section 4.5 states these results; the paragraphs it condensed are here.

Raw transfer is heterogeneous and includes anti-predictive di-
1760 **rections.** Raw target AUC spans 0.326 to 0.686 (Fig. A.4). Twelve of 20
directions are above chance with CI support. Six are *below* chance with CI
support: both directions of Manavgat to Bejís and of Manavgat to Muğla,
plus Bejís to Evia and Evia to Bejís — of which only Manavgat to Bejís sur-
vives frame equalisation. Two intervals span 0.5. Even the best raw transfer,
1765 Evia to Manavgat at 0.686, stays far below that target’s own within-region
thermal performance of 0.870. Across all directions the raw deficit against
the within-region reference is 0.184 to 0.592 AUC. That reference is a region-
level blocked estimate, and Section 4.3 shows it is not matched to a transfer
evaluation.

1770 Those counts belong to the 2-cell blocking of Table B9. At the more
conservative 10-cell blocking the same points give 9 above, 4 below and 7
uncertain, and no direction changes side of the chance line (Appendix A(v)).

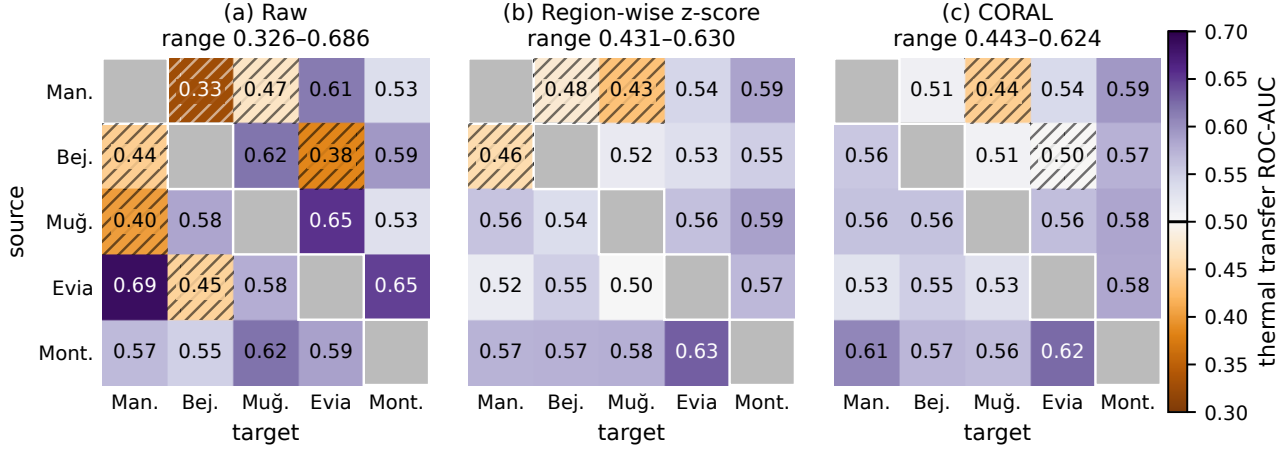


Figure A.4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. (a) Raw transfer, no adaptation. (b) After region-wise z-score standardisation. (c) After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table B9.

Four of the six below-chance directions keep their support there. Bejís to Manavgat and Manavgat to Muğla lose it and carry no verdict. The qualitative statement is unchanged. The counts should not be read as exact.

In precision terms the transfer is worse than the ROC figures suggest. A susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity. Across the twenty directions the thermal model’s PR-AUC averages **0.156 against a no-skill baseline of 0.136.** Six of the twenty fall below their own no-skill baseline at the point estimate, five of them with intervals entirely below it; the exception is Bejís to Manavgat, whose interval covers its baseline. Only one direction exceeds twice its baseline. These are frame-as-drawn quantities and the PR arm was not recomputed on the collar. Per-direction values are in Appendix B, Table B4.

The static baseline does not transfer either. The same twenty directions were run with the terrain, fuel and greenness baseline alone, giving a mean target AUC of **0.537 against 0.541** for the thermal model. Static attributes of a place are the class Dimarco et al. transfer successfully and the class this paper’s framing treats as portable; here that class is itself barely above chance. The transfer failure below is therefore not specific to the dynamic block: a baseline that does not travel, plus pre-fire thermal state, gives a model that does not travel.

The thermal block’s paired contribution to transfer, with its interval. Differencing the two matrices direction by direction gives a mean of **+0.004**, but the mean is not the informative statistic. The individual contributions span **−0.148 to +0.133**, twelve positive and eight negative, so the

spread is thirty times the mean and the sign belongs to the pair rather than to the block; a mean near zero records cancellation, not consistent absence of effect. The directions are not independent, since each region appears in eight of the twenty, so the interval depends on the resampling unit — **all four units the design permits give the same answer**, from $[-0.027, +0.034]$ treating directions as independent to $[-0.037, +0.046]$ jackknifing regions, and none propagates within-direction sampling variability (Appendix A(o)).

1805 The leave-one-region-out jackknife shows how little the mean is anchored: **dropping Evia alone reverses its sign.**

On the frames as drawn, six directions are below chance with interval support, the sharpest at 0.326 $[0.305, 0.349]$. Section 4.4 has shown most of that count is a property of the frames: equalising them leaves

1810 **one**, Manavgat to Bejís at 0.417 $[0.349, 0.488]$, supported at the 10 km collar though its interval covers chance at 5 km. That direction still needs a mechanism acting on the direction of the relationship, because no account of merely lost skill produces a reliably reversed ranking, and Section 4.6 pursues it. Everything below this point is computed on the frames as drawn

1815 and should be read against Section 4.4. Per-split and per-scar detail is in Appendix A(i); per-direction values for all twenty directions, raw and under both adaptations, are in Appendix B, Table B9.

Label-blind adaptation compresses the matrix toward chance rather than repairing it. Under region-wise z-scoring the twenty direc-

1820 tions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half the raw spread, with no adapted direction exceeding 0.631 against the unmatched within-region references of 0.859 to 0.918. Adaptation raises the

failing directions and degrades most of those that already transferred. Taking the better of the two adaptations per direction, 14 of the 20 end closer
1825 to chance than they began and 6 end further from it; five of those six involve Montiferru, the smallest and last-added region, and move upward, while the sixth is Manavgat→Muğla moving downward from 0.470 to 0.443. The 14 to 6 split should be read at the precision of limitation (ix) in Appendix C.5, since Bejís→Manavgat is counted as compressed on a margin of 0.001.

1830 The per-direction decomposition is in Appendix A(j), Table A5; two figures from it matter here and are used below.

Against the right reference, adaptation is not failing (Fig. A.5). The three controls above give an achievable reference for a model applied to a fire it has not seen: 0.574 for the half-split, 0.552 for the leave-one-scar-out.
1835 CORAL, committed to in advance, averages **0.552** across the twenty directions against 0.541 raw, which is exactly that reference. Taking whichever of the two adaptations scores higher per direction gives 0.556, but that selection uses the target labels the protocol forbids, so it is an oracle upper bound rather than an achievable result and is reported as one. What it does
1840 is regress the matrix towards that value: fourteen of twenty directions move closer to chance, which harms the directions that already worked and helps the ones that did not. The verdict that follows is narrower than “alignment fails”. Alignment cannot exceed what a model can achieve on an unseen fire, and it does not.

1845 In the six directions where raw transfer was below chance, the best label-free method recovers at most 34 % of the gap to the within-region reference, so the remaining unrecovered fraction is at least 0.66 everywhere. The half-

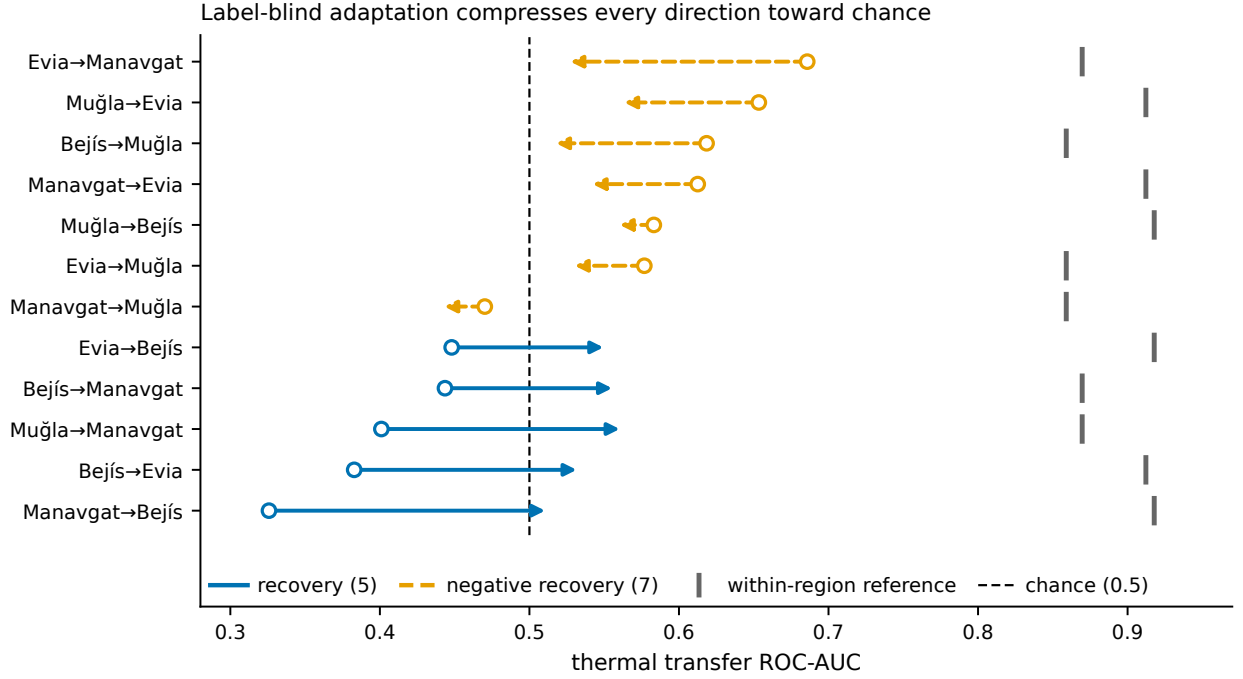


Figure A.5: **Label-blind adaptation regresses transfer onto the level a model reaches on an unseen scar, rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

split control above shows that the larger part of that remainder is already incurred inside the region, so it should not be read as a measure of concept shift. On the four-AOI twelve-direction subset for which the decomposition is defined (Appendix A(j)), seven directions show *negative* recovery, meaning adaptation moves the score away from the reference; in six of those raw transfer was already above chance and adaptation destroyed that advantage. Label-free alignment therefore does not act as a repair mechanism.

1855 *Appendix A.15. A(y). The diagnostics, elaborated*

Section 4.6 states these results; the paragraphs it condensed are here.

Twenty candidate diagnostics from five families were each rank-correlated with the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap.

1860 Table 5 of the body (Section 4.6) groups them by family and is not repeated here.

Geographic separation does not order the matrix on either construction: over all twenty directions the Spearman correlation between centroid separation and transfer is -0.32 , and on the twelve-direction common subset 1865 -0.24 , both spanning zero. The two nearest directions are among the worst on the frames as drawn, while the 2,802 km pair returns 0.326 and 0.444, so the transfer mean of 0.541 is not the value at any one separation.

Only two diagnostics have intervals excluding zero, and both are conditional: the sign-agreement fraction over interval-supported features 1870 at $\rho = +0.84$ [$+0.58$, $+0.88$], and its cosine variant at $+0.81$. No marginal measure was shown to order the matrix, including area-of-applicability dissimilarity, climatic distance and geographic distance, and neither were the

niche-overlap and regime families. The learned domain classifier is at ceiling, separating source from target at $\text{AUC} \geq 0.96$ for every pair; it always
1875 succeeds, which is why it carries no ordering information.

Four limits are stated with the result rather than after it. **Size:** the index’s tie structure caps the achievable Spearman at $+0.861$, so the observed $+0.840$ sits on that ceiling, and its exact one-sided permutation p of 0.0060 is the smallest that structure can produce against a Bonferroni threshold of
1880 0.0026 over nineteen computed variants — **no outcome of this diagnostic could have cleared family-wise correction on ten effective pairs.** **Labels:** signed associations need burned labels in both regions, so the family that appears to work is not available before deployment while the family that fails is. **Selection,** the sharpest of the three: the two rows that clear zero are
1885 the *supported-feature* variants, whose subset is chosen by whether two regions’ bootstrap intervals happen to be disjoint — a data-dependent selection made on the same data, with no correction. Their unselected counterparts over all nine features are $\rho = +0.50 [-0.17, +0.83]$ and $+0.18 [-0.40, +0.72]$, both spanning zero, so the result lives in the selection step and is reported as such.

1890 The families sit on unequal samples — twelve directions for the marginal, applicability, climatic and geographic rows, sixteen for the supported-conditional rows, twenty for the rest — so every row was recomputed on the common twelve. Published values reproduce to 4.8×10^{-5} , the conditional rows still lead at $+0.87 [+0.65, +0.88]$ and $+0.85 [+0.43, +0.88]$, and every marginal
1895 row still spans zero. The ordering is not an artefact of unequal samples.

A fourth limit, established in Section 4.4, removes the result entirely. The index is built from signed associations that Section 4.4 shows

to be frame artefacts, and was correlated against transfer measured on the same frames. Recomputed on an equalised frame it is unanimous, with no
1900 variance left to correlate. Everything in this section is what the original protocol yields; the conclusion that survives is that **no diagnostic tested here was shown to order transfer once the frames are comparable.**

Appendix A.16. $A(z)$. The four evaluations, elaborated

Section 4.3 states these results; the paragraphs it condensed are here.

1905 **Where the skill is lost, on a matched comparison.** Four evaluations are reported; the last three are scored on **identical cells**, so they differ only in what the model was trained on, and the first shows why an unmatched comparison misleads. The held-out unit is a burned connected component of at least 50 cells together with all cells within 2 km of it. What makes that a
1910 harder problem than a whole region is the composition of its negatives, not its burned fraction, and that can be shown rather than argued. Taking the same fitted model and the same out-of-fold predictions and scoring them on a random sample of region cells drawn at the scar area's own burned fraction gives **0.782 against the region-wide 0.782**: matching the prevalence
1915 changes nothing, at -0.000 $[-0.003, +0.002]$ over the nine scars. Scoring the same predictions on the scar area itself gives 0.627 , a fall of **+0.155** **[+0.093, +0.217]**. The whole effect is the negative pool. Every negative in a scar collar is fire-adjacent, sharing the terrain, land cover and synoptic conditions of the positives, whereas a region's negatives include its easy far
1920 field. The burned fraction, 34 to 87 % against 3.8 to 28.7 % for a region, is a symptom of that construction, and ROC-AUC is in any case invariant to class balance at fixed class-conditional distributions.

Table 2 of the body (Section 4.3) carries the four evaluations and is not repeated here; its rows are labelled A to D and referred to by those letters
1925 below.

Table 2's four rows are means over the **same eight scars**. A ninth burned component, Bejís, is excluded **from it**: it is that region's only component of any size, so holding it out leaves no usable source model and there is no row C for it. It is *not* excluded from the arms that need no leave-one-scar-out
1930 model, so the prevalence control two paragraphs above and Tables A3 and A4 are computed on nine scars. That is why their means, 0.782 and 0.627, differ from Table 2's 0.776 and 0.634. Rows A, B and D are given there on the eight so that the differences are paired, and the intervals are Student *t* over those eight, which is the resampling unit for this arm rather than the
1935 spatial-block bootstrap used elsewhere in the paper. Eight is a small number and the intervals are wide accordingly. Four of the eight scars are in Muğla and two in Montiferru, so they are not independent; row A in particular is a region-level quantity repeated across the scars of a region, and its interval is pseudo-replicated and should not be read as coverage. The two differences
1940 are computed per scar and paired, which is what they are reported for.

The same model, scored two ways on the same region, differs by 0.143 AUC. Rows A and B are one model. The only change is which cells it is scored on: the whole region, or the burn scar and its 2 km collar. That change alone costs **0.143** [**+0.077**, **+0.208**] of the 0.225 [+0.157, +0.293]
1945 fall from A to C, about two thirds, and it is the size of the entire increment this literature usually reports. Both intervals are Student *t* over the eight scars, so the “two thirds” is a ratio of two estimates and is compatible with

anything from roughly a third to nine tenths; the point of the comparison is the size of the numerator, not the precision of the fraction. A region-wide
1950 blocked figure is therefore an upper bound on what the same model achieves where the fire actually is, and the difference is not small enough to ignore. Any comparison of a scar-level result against a region-level reference inherits that, and the region-level reference is what a paper of this kind normally reports.

1955 **Neither withholding the fire nor moving 2,800 km has a measurable cost.** B minus C, the fire-specific residual on identical cells, is **+0.082** **[−0.011, +0.175]**, which spans zero. C minus D, the effect of replacing a same-region model with one fitted between 306 and 2,802 km away, is **−0.003** **[−0.075, +0.069]**, which also spans zero and is an order of mag-
1960 nitude smaller. The second is the cleaner result: on the same cells, a model fitted in another country does as well as one fitted in the same region with the scar withheld, and it survives restriction to Muğla, the only region where holding out one scar still leaves the source model properly trained, at 0.579 against 0.597.

1965 **Row D averages four foreign models over a large spread,** 32 scar-by-source combinations running 0.374 to 0.724 with ten below chance, so the pooled 0.555 says the *average* foreign model matches a same-region one, not that any particular one does. One confound must be stated: holding out a region’s only large scar also removes most of its positives, so row C
1970 mixes “the fire was withheld” with “almost all the positives were withheld”; restricted to Muğla, where four scars remain, C is 0.579 and D is 0.597 and the comparison still shows nothing. **With eight scars, three of them**

starved, this design cannot establish a fire-specific residual, only bound it at about 0.18. Rows C and D both sit close to chance, so the comparison between them has little dynamic range: what is measurable is the change of evaluation geometry between rows A and B. **It is not shown to be zero on an unseen fire, and eight scars cannot show that; what this design establishes is that it is not established there.** Nor does it establish the fire event as the unit, because the held-out patch is defined by the labels and its identity cannot be separated from its location. Sweeping the two parameters that define that patch — minimum burned-component size over 25 to 200 cells and connectivity over the 4- and 8-neighbourhoods — moves the row-C mean only between 0.545 and 0.566, and it is flat across buffers of 2, 5 and 10 km. Per-scar and per-source values, the full sweep and the half-split control are in Appendix A(i).

The increment declines with the holdout, and is not established once the fire is withheld. Contribution 1 is about the paired thermal-minus-baseline difference, so the same evaluations were run on it.

Evaluation	Thermal increment	Interval
Blocked cross-validation, per region	+0.056 to +0.153	every interval above zero
Within-region half-split	+0.027	positive in 13 of 18 splits
Leave-one-scar-out	+0.022	[−0.032, +0.077]
Cross-region, twenty directions	+0.004	[−0.028, +0.036]

The point estimate falls monotonically as the holdout hardens and the last two intervals span zero, so **the within-region increment of +0.056 to**

+0.153 is substantially a property of interleaved holdout. Per-scar values are in Appendix A(i).

Appendix A.17. A(aa). The six further arms, summarised

1995 Six further arms bear on the findings above without changing them; each is reported in full in Appendix A. **The contrast pair** (A(s)): the most environmentally similar pair in the matrix is among the weakest in transfer and the least similar among the stronger, so high envelope overlap does not buy transfer — an ordinal claim, not one about the endpoints. **Interven-**
2000 **tions** (A(n)): pooling four regions into one training set does not recover what single-source transfer loses, and removing the two reversing predictors costs -0.081 of within-region AUC while returning $+0.014$ $[-0.017, +0.045]$ on transfer, so a local cost is measured and no compensating transfer gain is. **Sensitivity** (A(a)–A(h)): eight design choices were varied with everything
2005 else held fixed; none changes a conclusion above, except arm (f), which tests a claim of Section 1.2 and does not uphold it. **The same geography, a second fire** (A(m)): the one arm that held place fixed shows an elevation reversal on the frame as drawn, which Section 4.4 withdraws. **Distance** (A(t)): within a single region the model is already at chance by 10 to 20 km
2010 from its training cells, which bounds how far a surface of this kind can be carried but cannot be turned into an attribution, for the reason given there. **The price of labels** (A(u)): thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s matched ceiling in three of six directions and 30 to 57 % in the rest, which is 7 to 20 % of the target’s natural-vegetation population
2015 and not a cheap answer. Two limits bound it: at the top budget the selection pool is nearly exhausted, so the narrow upper-budget intervals reflect sat-

uration rather than precision, and **the labelled blocks are drawn from the event being predicted, which is not a resource available before that event burns.**

2020 **Appendix B. Appendix B. All twenty transferability diagnostics**

Section 4.6 summarises this table by family. The full ranking is given here because the negative result is the point: a reader should be able to see every candidate that was tried, not only the families' best members, and to check that no diagnostic was dropped after it failed.

2025 **Table B1. All transferability diagnostics versus raw thermal transfer (20 ordered directions).** Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with $n = 12$ exist only for the four-AOI subset, because those diagnostics were never produced for Montiferru. The supported-features conditional rows use the 16 directions (8 pairs) with
2030 at least one CI-supported feature.

Diagnostic	Family	Exp.	n	dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y x) conditional	+	16		+0.84 [+0.58, +0.88]	yes
Cosine, supported features	P(y x) conditional	+	16		+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	P(y x) conditional	+	20		+0.50 [−0.17, +0.83]	no
Vector	P(y x)	+	20		+0.27 [−0.36, +0.77]	no
Spearman, all 9	conditional					
Agreement count, all 9	P(y x) conditional	+	20		+0.18 [−0.40, +0.72]	no
Schoener’s D, 1-D mean	P(x y=1) niche	+	20		+0.24 [−0.45, +0.74]	no
Warren’s I, 1-D mean	P(x y=1) niche	+	20		+0.22 [−0.42, +0.73]	no
Schoener’s D, PCA-2D	P(x y=1) niche	+	20		+0.10 [−0.51, +0.68]	no
Warren’s I, PCA-2D	P(x y=1) niche	+	20		−0.07 [−0.66, +0.49]	no
Mahalanobis, burned centroids	P(x y=1) niche	−	20		−0.23 [−0.75, +0.44]	no
Domain-classifier AUC	P(x) marginal	−	20		−0.32 [−0.78, +0.33]	no
Predictor-space mean dissimilarity	P(x) marginal	−	12		−0.10 [−0.54, +0.43]	no
Predictor-space p95 dissimilarity	P(x) marginal	−	103	12	−0.08 [−0.59, +0.49]	no
Fraction inside weighted AoA	P(x) marginal	+	12		+0.22 [−0.48, +0.59]	no
Fraction inside unweighted	P(x) marginal	+	12		+0.08 [−0.89, +0.63]	no

Table note. A null row means the diagnostic was **not shown to order transfer** on this design, not that it was shown incapable of ordering it. With ten effective region pairs the power is low, and the intervals are wide
2035 enough to admit moderate true correlations in either direction. The last row is retained rather than deleted because it was computed: on two directions the statistic has no meaningful value, and reporting that is more honest than dropping the variant.

Reading the two rows that clear zero. Both are supported-feature
2040 variants, where the predictor subset is chosen by whether two regions' bootstrap intervals happen to be disjoint. That is a data-dependent selection made on the same data, uncorrected. Their all-nine-feature counterparts are in the table and both span zero. The result lives in the selection step, and Section 4.6 says so.

Equal-sample check. The families sit on unequal samples: marginal,
2045 applicability, climatic and geographic rows on twelve directions, the supported-conditional rows on sixteen, the rest on twenty. Recomputing every row on the common twelve directions reproduces the published values to 4.8×10^{-5} and leaves the ordering unchanged — the conditional rows still lead at +0.87
2050 [+0.65, +0.88] and +0.85 [+0.43, +0.88], every marginal row still spans zero. Source: `paper/diagnostics_common_subset.md`.

Appendix B.1. B2. The signed univariate associations the reversal claim rests on

Appendix A(n) drops two predictors because their signed association with
2055 burning reverses between regions with bootstrap support on the frames as drawn. Section 4.4 shows that both supported elevation reversals disappear

once the frames are equalised, so this table is the evidence for the feature-removal arm and for the narrowed mechanism claim inside Contribution 1, not for a general reversal mechanism. It was computed for the frozen analysis and is reproduced here because the claim is otherwise asserted rather than shown.

Table B2. Signed univariate AUC of each predictor against burned, by region. Primary natural-vegetation population; 10-cell (~ 5 km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 means lower values rank burned and is a direction rather than weakness. **Bold** marks a region whose own interval excludes 0.5.

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
elevation_mean	0.374 [0.289, 0.471]	0.643 [0.558, 0.729]	0.611 [0.532, 0.690]	0.541 [0.448, 0.626]	0.584 [0.395, 0.762]
slope_mean	0.531 [0.423, 0.642]	0.521 [0.439, 0.605]	0.637 [0.582, 0.686]	0.487 [0.418, 0.554]	0.652 [0.506, 0.771]
ndvi_mean	0.636 [0.587, 0.676]	0.559 [0.497, 0.619]	0.662 [0.616, 0.704]	0.639 [0.575, 0.701]	0.586 [0.450, 0.704]
lst_anomaly_mean	0.482 [0.428, 0.530]	0.418 [0.364, 0.480]	0.485 [0.395, 0.566]	0.640 [0.567, 0.710]	0.395 [0.285, 0.535]
current_lst_mean	0.538 [0.452, 0.621]	0.477 [0.401, 0.547]	0.325 [0.271, 0.382]	0.377 [0.301, 0.456]	0.370 [0.248, 0.513]
current_tvdi_mean	0.552 [0.460, 0.641]	0.517 [0.429, 0.595]	0.336 [0.275, 0.398]	0.362 [0.285, 0.442]	0.356 [0.233, 0.499]
tvdi_difference_mean	0.449 [0.391, 0.505]	0.512 [0.443, 0.583]	0.490 [0.396, 0.575]	0.519 [0.444, 0.589]	0.378 [0.282, 0.497]
downscaled_lst_mean	0.552 [0.466, 0.637]	0.484 [0.400, 0.560]	0.307 [0.253, 0.366]	0.377 [0.297, 0.459]	0.365 [0.240, 0.511]
fused_lst_mean	0.540 [0.454, 0.622]	0.481 [0.404, 0.551]	0.325 [0.272, 0.383]	0.376 [0.300, 0.456]	0.370 [0.248, 0.513]

What counts as a reversal. A pair of regions is called a reversal

only when their signed associations point to opposite sides of 0.5 **and each region’s own interval excludes 0.5**. That is stricter than requiring the two regions’ intervals to be disjoint, and the difference matters: for `current_1st_mean` between Manavgat and Muğla the two intervals are disjoint, but Manavgat’s own interval, [0.452, 0.621], includes 0.5, so Manavgat has no established direction to reverse from. That pair is a point reversal, not a supported one.

Table B3. The cross-region reversals that meet the stricter criterion. Difference intervals are from the same paired bootstrap.

Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
<code>elevation_mean</code>	Manavgat	0.374	Bejís	0.643	+0.269 [+0.138, +0.397]
<code>elevation_mean</code>	Manavgat	0.374	Muğla	0.611	+0.235 [+0.102, +0.360]
<code>1st_anomaly_mean</code>	Bejís	0.418	Evia	0.640	+0.221 [+0.123, +0.313]

Three pair-level reversals across **two** features, which is why Section 3.12 removes exactly those two. Twenty-nine further pairs reverse at the point estimate only, spread across eight of the nine features, and they are not counted. The conservative criterion costs the paper findings rather than manufacturing them: a difference interval on the pair, which is the instrument Appendix A(m) uses, would support more reversals than the three listed here.

Appendix B.2. B4. The transfer matrix in precision-recall terms

ROC-AUC is reported throughout the main text for comparability with the susceptibility literature. A susceptibility surface is used as a ranked area budget, so precision-recall is the operational quantity, and at target
2090 prevalences of 3.8 to 28.7 % the two can differ sharply. Read from the same frozen step9b exports as Table B9.

Table B4. Thermal transfer, PR-AUC against the no-skill baseline. The baseline is the target's burned prevalence. Lift is PR-AUC divided by that baseline; a lift of 1 is no better than random ranking. Ordered by
2095 lift.

Direction	ROC-AUC	PR-AUC	No-skill	Lift
Evia $\rightarrow$ Manavgat	0.686	0.094	0.038	2.45
Bejís $\rightarrow$ Montiferru	0.594	0.289	0.212	1.36
Evia $\rightarrow$ Montiferru	0.647	0.283	0.212	1.34
Bejís $\rightarrow$ Muğla	0.619	0.093	0.070	1.33
Muğla $\rightarrow$ Evia	0.653	0.379	0.287	1.32
Montiferru $\rightarrow$ Bejís	0.548	0.093	0.072	1.28
Montiferru $\rightarrow$ Muğla	0.619	0.089	0.070	1.27
Evia $\rightarrow$ Muğla	0.577	0.086	0.070	1.23
Muğla $\rightarrow$ Bejís	0.583	0.088	0.072	1.22
Manavgat $\rightarrow$ Evia	0.613	0.343	0.287	1.20
Manavgat $\rightarrow$ Montiferru	0.533	0.243	0.212	1.15
Montiferru $\rightarrow$ Manavgat	0.567	0.043	0.038	1.11
Montiferru $\rightarrow$ Evia	0.586	0.317	0.287	1.11
Muğla $\rightarrow$ Montiferru	0.531	0.214	0.212	1.01
Bejís $\rightarrow$ Manavgat	0.444	0.034	0.038	0.90
Manavgat $\rightarrow$ Muğla	0.470	0.063	0.070	0.90
Evia $\rightarrow$ Bejís	0.448	0.059	0.072	0.82
Bejís $\rightarrow$ Evia	0.383	0.226	0.287	0.79
Muğla $\rightarrow$ Manavgat	0.401	0.029	0.038	0.75
Manavgat $\rightarrow$ Bejís	0.326	0.049	0.072	0.67
Mean	0.541	0.156	0.136	1.16

Six directions fall below their own no-skill baseline, and only one exceeds twice it. The six are the same six that are below chance on ROC-AUC, which is what a reversed ranking predicts in either metric. Section 4.4 shows that this count is largely a property of the evaluation frames rather than of

2100

a reversed predictor-burning relationship.

Appendix B.3. B5. The same-geography event pair, in full

Appendix A(m) reports this arm and Section 4.4 withdraws its **elevation** reversal as a frame artefact; Appendix A(o) explains why the thermal
2105 channels of this arm carry no verdict either way. The per-feature values are kept here because the arm is what motivated the frame test, and because its structural asymmetries have no analogue in the twenty-direction matrix.

Table B5. Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against burned, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42
2110 (Section 3.15). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm.** Table 1’s note sets sixteen as the floor this design supports at that blocking, so the 2022 intervals here fall
2115 below the paper’s own standard and are read as indicative, exactly as the 20-cell row of Table 1 is. The 2022 arm is additionally a single compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

Appendix B.4. B6. The evaluation frames, and the signed associations they produce

Section 4.4 states these results; the per-region values are here.

Table B6. Evaluation-frame geometry of the five study regions.

Primary natural-vegetation population. Distance is Euclidean to the nearest burned cell on the 500 m grid, at 0.45 km per cell. Source `aoi_frame_auc.csv`; recomputable by `paper/code/verify_aoi_frame.py`.

Region	cells	burned	median distance to burned	share beyond 10 km
Manavgat	20,511	784	13.4 km	60.1 %
Bejís	15,190	1,100	13.5 km	63.1 %
Muğla	41,730	2,911	11.3 km	55.3 %
Evia	9,298	2,664	8.0 km	43.7 %
Montiferru	2,544	539	2.7 km	2.1 %

Table B7. Signed univariate AUC, frame as drawn against a 10 km collar. Point estimates; the intervals that decide the reversal question are given in the text below and in `collar_frame_bootstrap.csv` (10-cell blocks, 1000 replicates, seed 42). Signed and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is a direction, not weakness. The collar drops no burned cells in any region.

Signed univariate AUC	Manavgat	Bejís	Muğla	Evia	Montiferru	straddles 0.5
elevation, full frame (Table B2)	0.374	0.643	0.611	0.541	0.584	yes
elevation, 10 km collar	0.561	0.614	0.606	0.648	0.581	no
current LST, full frame	0.538	0.477	0.325	0.377	0.370	yes
current LST, 10 km collar	0.386	0.405	0.332	0.286	0.376	no
current TVDI, full frame	0.552	0.517	0.336	0.362	0.356	yes
current TVDI, 10 km collar	0.392	0.454	0.342	0.250	0.361	no

Table B8. Region summary: populations and gate outcomes. Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled pop-
2135
2140 ulation, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural- veg frac- tion	Grade verdict
Manavgat 2021	24,150	24,087	796	0.033	20,511	784	0.038	0.984	pass
Bejís 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Mugla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (ex- tended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Monte Lezvi 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

Table B9. Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs

(1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.461]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

Table B10. The most and least environmentally similar pairs, on both frames. Schoener’s D is computed over burned cells only and is therefore collar-invariant. Transfer values are the two ordered directions of each pair; ranks are out of the twenty directions on the equalised frame. As-drawn transfer is read from Table B9, collar transfer from `aoi_frame_transfer.csv`.

	Manavgat–Muğla	Bejís–Montiferru
Schoener’s D , mean 1-D	0.826 (highest)	0.479 (lowest)
per-feature D	0.77 to 0.89	0.23 to 0.77
transfer, frames as drawn	0.470, 0.401	0.594, 0.548
transfer, 10 km collar	0.551, 0.510	0.669, 0.624
rank of 20 on the collar, from the bottom	5th, 2nd	15th, 11th
target cells inside the AoA	0.875, 0.531	—

2155 Appendix C. Appendix C. Protocol detail

These protocols are given here in full rather than in Methods, because each is a specification a reader needs only when checking the corresponding result, and none is needed to follow the argument. Sections 3.2, 3.3, 3.11 and 3.15 state in summary what each does.

2160 Appendix C.1. C.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate

Labels come from MODIS MCD64A1 Collection 6.1 (MODIS/061/MCD64A1) retrieved through Google Earth Engine (Gorelick et al., 2017), whose omission and commission characteristics (Boschetti et al., 2019) bound every
2165 model here. The product is exported onto the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which duplicates each native ~ 500 m observation across the 30 m pixels beneath it.

The analysis grid is therefore **reconstructed rather than native**: the 30 m grid is partitioned into non-overlapping blocks of $\text{round}(500 / 30) =$
2170 17×17 pixels, giving a nominal cell edge of 510 m, and each cell is identified

by its integer block indices `row_500m` and `col_500m`. This approximates the MODIS sinusoidal cell rather than reproducing it, being anchored to the Landsat/EPSG:4326 reference grid, and marginal blocks are truncated. The cell is square in degrees but not on the ground, about 510 m north to south and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against a MODIS cell's 0.250 km²; every block-size label in this paper is therefore the north-south dimension. The companion paper develops what follows from that.

A cell's representative burn date is the **mode** of the positive sub-pixel day-of-year values in the block, tested against the region's label window. Because the exported raster carries no out-of-window positives, in this dataset a single in-window positive sub-pixel makes the cell burned. The fraction of positive sub-pixels agreeing with the modal date is recorded as `burn_date_pixel_agreement_fraction` but no agreement threshold is imposed. The label never affects a cell's eligibility for modelling: unburned, all-no-data and out-of-window cells all remain as the negative class.

Two safeguards are recorded rather than assumed. Cells that burned inside a region's own predictor window are removed from its analysis universe rather than counted as unburned, which ran for Muğla, Evia and Montiferru, is recorded as not run for Manavgat and has no recorded status for Bejís. Burning in earlier years is screened for no region in the five-region cohort; the only historical exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.15.

Before any modelling each region passes a gate that answers one question: of the cells labelled burned, what fraction is dominated by natural

vegetation? A region is admitted as a wildfire candidate when that fraction reaches 0.50 and at least 30 burned cells are present, and is rejected as a cropland-dominated control when the cropland fraction reaches 0.50 instead. The gate uses ESA WorldCover classes aggregated to the same reconstructed
2200 cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

Appendix C.2. C.2 Transferability diagnostics versus transfer

Twenty candidate diagnostics from five families are each rank-correlated
2205 against the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-response direction $P(y|x)$.

2210 The **marginal** family includes area-of-applicability-style dissimilarity in scaled, importance-weighted predictor space (Meyer and Pebesma, 2021, 2022; Ludwig et al., 2023), climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener’s D (Schoener, 1968) and Warren’s I (Warren et al., 2008) over the burned cells of each region. The **regime** family includes burned-patch component counts and effective component counts. The **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are in-
2220 dividually interval-supported.

Two properties of this design are stated in advance because they bound what it can show. The candidate set was fixed in a project analysis log before the correlations were computed; that log is not a formal pre-registration and no independent timestamped registration exists. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are evaluated at that power with no family-wise error control claimed; where a diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

Appendix C.3. C.3 Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla is the one region for which a second fire event is analysed on the identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening on it, matching the 2021 durations exactly.

This is not a clean temporal-transfer design and is not presented as one. The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no difference can be attributed to the year alone. The registered claim is same-geography event-to-event. The pair is deliberately kept out of the twenty-direction matrix, and that exclusion is enforced in released code rather than merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion applies to the 2022 arm alone, masking every cell that burned in 2021 so the

previous year’s scar does not enter the following year’s analysis: it removes 3,073 cells, of which 2,941 belong to the primary population, taking it from 41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of 38,790 cells and three byte-identical static predictors, and in the 2022-to-2021 direction every target positive lies outside the source training population. Appendix A(m) states what follows for how these two numbers may be read.

Table C1. Study regions, areas of interest and temporal windows. Bounding boxes are in EPSG:4326, as registered. Window lengths in brackets are inclusive day counts. The baseline years are the four window-symmetric years preceding each predictor window.

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Manavgat 2021 (Türkiye)	31.05, 36.72, 31.85, 37.35	2021-06-01 to 2021-07-27 (57 d)	2021-07-28 to 2021-08-31 (35 d)	2017, 2018, 2019, 2020
Bejís 2022 (Spain)	-1.05, 39.68, -0.35, 40.15	2022-06-15 to 2022-08-14 (61 d)	2022-08-15 to 2022-09-30 (47 d)	2018, 2019, 2020, 2021
Muğla 2021 (Türkiye)	27.1, 36.6, 28.9, 37.45	2021-06-01 to 2021-07-28 (58 d)	2021-07-29 to 2021-09-15 (49 d)	2017, 2018, 2019, 2020
North Evia 2021 (Greece)	23.05, 38.55, 23.85, 39.15	2021-06-05 to 2021-08-02 (59 d)	2021-08-03 to 2021-09-30 (59 d)	2017, 2018, 2019, 2020
Montiferru 2021 (Italy)	8.45, 40.05, 8.75, 40.27	2021-05-25 to 2021-07-23 (60 d)	2021-07-24 to 2021-08-31 (39 d)	2017, 2018, 2019, 2020

Appendix C.4. C.4 Predictor provenance, compositing and the downscaler

Ten predictors are used, four baseline and six thermal, all summarised per cell over the predictor window, through the chain shown in Fig. C.1. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL

band with the water bit deliberately preserved; the coarse-resolution thermal input is MODIS/061/MOD11A1.

- **NDVI**, the predictor-window median of Landsat surface reflectance, is
2265 the baseline’s one time-varying member.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30.
- **Land cover** is ESA WorldCover v200 (Zanaga et al., 2022), entering
as the dominant class code.
- **Current LST**, the predictor-window median of Landsat surface tem-
2270 perature in °C.
- **LST anomaly**, a z-score of the current-window LST median against
the four baseline years, set to no-data where the baseline standard
deviation is below 1.0 °C or observations are too few.
- **TVDI** and **TVDI difference**. The Temperature-Vegetation Dryness
2275 Index (Sandholt et al., 2002) is computed in the LST-NDVI feature
space, with wet and dry edges taken as the 2nd and 98th LST per-
centiles within each of twenty NDVI bins and the index clamped to
[0, 1]. The edges are percentiles of the LST values a given scene con-
tains, so they are fitted per AOI and per window, and a TVDI of 0.5
2280 denotes a different physical moisture state in each region. The compan-
ion paper reports what that scene dependence does and does not ex-
plain. **TVDI difference** is the raw anomaly of that index against the
same four window-symmetric baseline years used for the LST anomaly:

Section numbers refer to *Methods*.

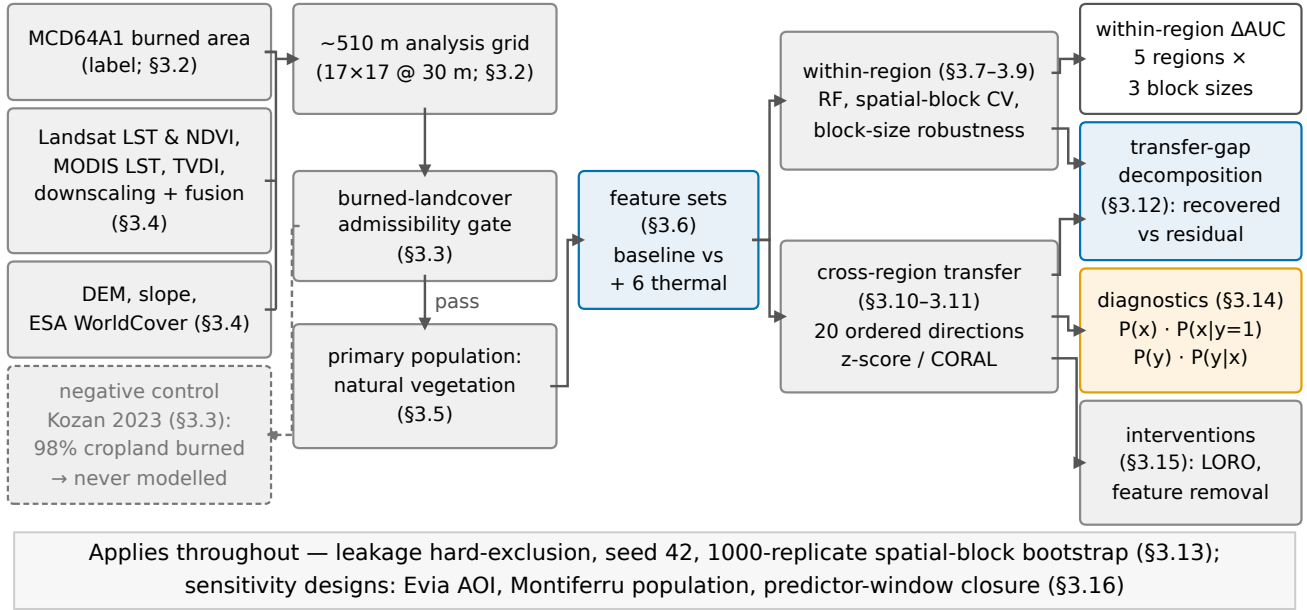


Figure C.1: **Processing pipeline and evaluation programme.** Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

2285 $\text{tvdi_difference} = \text{current_tvdi} - \text{mean}(\text{baseline_tvdi})$, in index units rather than standard deviations, which is why it is reported alongside the z-scored channel rather than in place of it.

• **Downscaled LST and fused LST.** A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30 m grid. The fused product equals observed Landsat LST wherever that is valid, and the downscaled surface only where it is not. Gap-filling therefore never replaces or blends a valid observation. The gap-filled share is 0.11 % to 9.70 % by region. The downscaler’s own inputs include coordinates, which is the one route by which a coordinate-derived surface re-enters a feature set from which Section 3.14 excludes coordinates. Appendix A(g) reports the increment without these two channels.

Appendix C.5. C.5 Limitations, in full

Section 5.8 states the four that bind the conclusions; all eleven are here.

(i) **No meteorological covariates** enter the models, so we cannot say how local skill and portability behave for a mixed thermal-plus-weather predictor set.

(ii) **The same-geography comparison covers one region only**, and even there year and seasonal phase are confounded. That confound cannot be resolved in this study area, and the reason is specific: the two events sit 42 days apart in median burn day-of-year, neither year contains a second event at the other’s phase, and a calendar-matched arm would carry nine burned cells against this design’s gate minimum of thirty. The positive-block

count for the 2022 arm is separately below the floor this design sets itself, at eleven against sixteen (Appendix A(m)). Its 331 burned cells also leave
2310 the thermal reversals unresolved at interval level, and the pair holds place fixed but not population. Those two arms are additionally the only transfer directions here computed by us rather than read from the pipeline author’s frozen export, with his unmodified code and the same pinned environment.

(iii) **All labels derive from a single burned-area product**, MCD64A1
2315 (Giglio et al., 2018), whose omission and commission characteristics (Boschetti et al., 2019) bound every model evaluated here. No second burned-area product covers 2021 and 2022 at this resolution; the companion paper reports what an independent active-fire observation says about the omission concern.

2320 (iv) **Evia remains the most imbalance-atypical population** even after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072 in three of the others.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires
2325 multi-year labels.

(vi) **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

2330 (vii) **The diagnostic correlations rest on an effective sample of ten region pairs**. Both the successes and the failures of Section 4.6 should be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained.**

It is where the conditional diagnosis bites hardest and where feature removal
2335 recovers most. Three candidates have now been tested and none survives:
its meteorology, which was not extreme; the quality screening of its coarse
thermal input, which propagates widely but moves no signed association by
more than +0.0003 (Appendix A(v), Appendix A(e)); and the evaluation
frame, which explains its elevation figure but not its transfer behaviour (Sec-
2340 tion 4.4). With one fire season per region the remaining candidates are not
separable in this design.

(ix) **The interval-support counts are less stable than the point
estimates behind them.** Several verdicts sit within a thousandth of their
reference value, and at 1 km blocking the published split of ten positive, seven
2345 negative and three uncertain turns on a lower bound of -0.00045 . The point
estimates and the sign pattern are stable; the counts are not. Every sentence
in this paper that leans on an exact count of supported directions should be
read at that precision. (x) **The five areas of interest are not comparable
frames, and this cohort cannot fully repair it** (Section 4.4). We report
2350 the equalised arm alongside the frame-as-drawn arm rather than replacing
one with the other, because the collar radius is itself a choice and 5 km and 10
km do not agree exactly (0.608 against 0.616). The deeper limitation is that
the frames were fixed upstream of this work, in `repo/`, so we can restrict
them but not extend them; a region whose rectangle is already fire-scale,
2355 Montiferru, cannot be given a far field for symmetry. Any future cohort
should fix the frame by an explicit accessible-area rule (Barve et al., 2011)
before any predictor is computed, and we treat that as the main design lesson

of this paper.

(xi) **One classifier family.** The headline numbers use a random forest
2360 with unlimited depth, the configuration most able to encode local structure
and least able to extrapolate. Appendix A(h) shows the transfer result is not
an artefact of that choice: three further estimators, including a penalised
linear one, all land between 0.510 and 0.556 and all place fourteen of twenty
directions above chance. Those are point estimates without intervals, so the
2365 ordering among them is not claimed as a result. Other model families were
not tried, and a different inductive bias might behave differently, but within
this family the negative result is a property of the predictors rather than of
an unregularised estimator.

Appendix C.6. C.6 Leakage control and reproducibility, in full

2370 An explicit forbidden-column set is enforced at every model fit. Coordinates
(lon, lat, row, col and their normalised forms), every burn-date and
label-provenance column, and the agreement fraction are excluded from all
feature sets, and the enforcement runs as an assertion rather than a con-
vention. The natural-vegetation mask is used only to define the population,
2375 never as a predictor. Spatial blocking prevents a cell from sharing a fold with
its own neighbours.

Reproducibility. All randomness uses seed 42 and the bootstrap uses
1000 replicates throughout. The transfer and adaptation analysis runs in
an environment separate from the upstream pipeline's. Every within-region
2380 model was therefore refitted there and compared against the frozen up-
stream output, and the independently implemented adaptation was com-
pared against the pipeline's own. The within-region comparisons agree ex-

2385 actly and the twenty transfer directions to within 1.6×10^{-7} . All numbers
 here were produced under scikit-learn 1.9.0 or verified against it; the im-
 plementation tolerance that applies if the version is not fixed is stated in
 Appendix C.5(vi), and the companion paper reports the version sensitivity
 and the reproduction check in full.

2390 **Sensitivity analyses.** Every headline result is repeated across two anal-
 ysis populations, three spatial-block sizes, the CORAL sweep, both feature
 sets and four classifier capacities; where a conclusion depends on one of
 those choices the dependence is reported rather than resolved by choosing
 the favourable setting (Appendix A).

*Appendix C.7. C.7 Transfer-gap decomposition and the concept-shift diag-
 nostic, in full*

2395 For each direction the gap between the target’s own within-region skill
 and the raw transfer result is split in two. One part is what the best label-
 free adaptation recovers, and the other is what it does not. The recovered
 fraction is $(\text{adapted} - \text{raw}) / (\text{within} - \text{raw})$, signed and unclipped, with
 its interval from the same paired bootstrap; it bounds what covariate-level
 2400 correction can achieve. Section 4.3 shows the remainder should not be read
 as a conditional residual, because much of it is incurred inside a single region
 (Appendix A(j)).

The mechanism is diagnosed by **signed univariate association**. For
 each numeric predictor the raw ROC-AUC of that predictor against **burned**
 2405 is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a
 value below 0.5 is read as a direction rather than as weakness. A reversal is
 called bootstrap-supported only when the two regions’ point estimates fall on

opposite sides of 0.5 and each region’s own interval excludes 0.5, under the same 10-cell spatial-block bootstrap. That is stricter than requiring the
2410 two regions’ intervals to be disjoint: a feature whose intervals are disjoint but one of which straddles 0.5 has not been shown to point anywhere in that region, so it is recorded as a point reversal only. Appendix B states the rule again beside the counts, and `conditional_similarity_transfer.json` carries it as machine-readable metadata.

2415 **Declarations**

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Competing interests. The authors declare no competing interests.

Ethics approval. Not applicable. This study involved no human participants, animal subjects, or personally identifiable data.
2430

Data availability. All satellite inputs are public. Burned-area labels

are MODIS MCD64A1 Collection 6.1 and land cover is ESA WorldCover, both retrieved through Google Earth Engine; no proprietary or restricted data were used. The frozen numeric outputs behind every table and figure, together with the analysis code that reads them, are available at <https://github.com/dryuemco/thermal-twin>. The upstream processing pipeline that produced those outputs is a separate release, <https://github.com/emrehann17/satellite-thermal-digital-twin>, and the commit of record for every number reported here is 48b56e7. No digital object identifier is minted and no archival deposit exists.

Code availability. The analysis code, its configuration and the scripts that regenerate each reported artefact are in the first repository above. Three components of the upstream pipeline fall outside the released commit, and a reader cannot re-run them: the reproduction-check driver and its validation package, on which Section 3.14 rests; the few-shot commit 19d825b, cited for provenance in the supplementary material; and the ERA5-Land diagnostic, whose manifest records commit a07ea33. Every other quantity in the paper can be regenerated from the released code and the frozen outputs.

Author contributions. Stated in CRediT terms. **Emrehan Metin:** Software, Data curation, Investigation, Validation, Writing – review and editing. **Yunus Emre Cogurcu:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Supervision.

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